

BFS Traversal

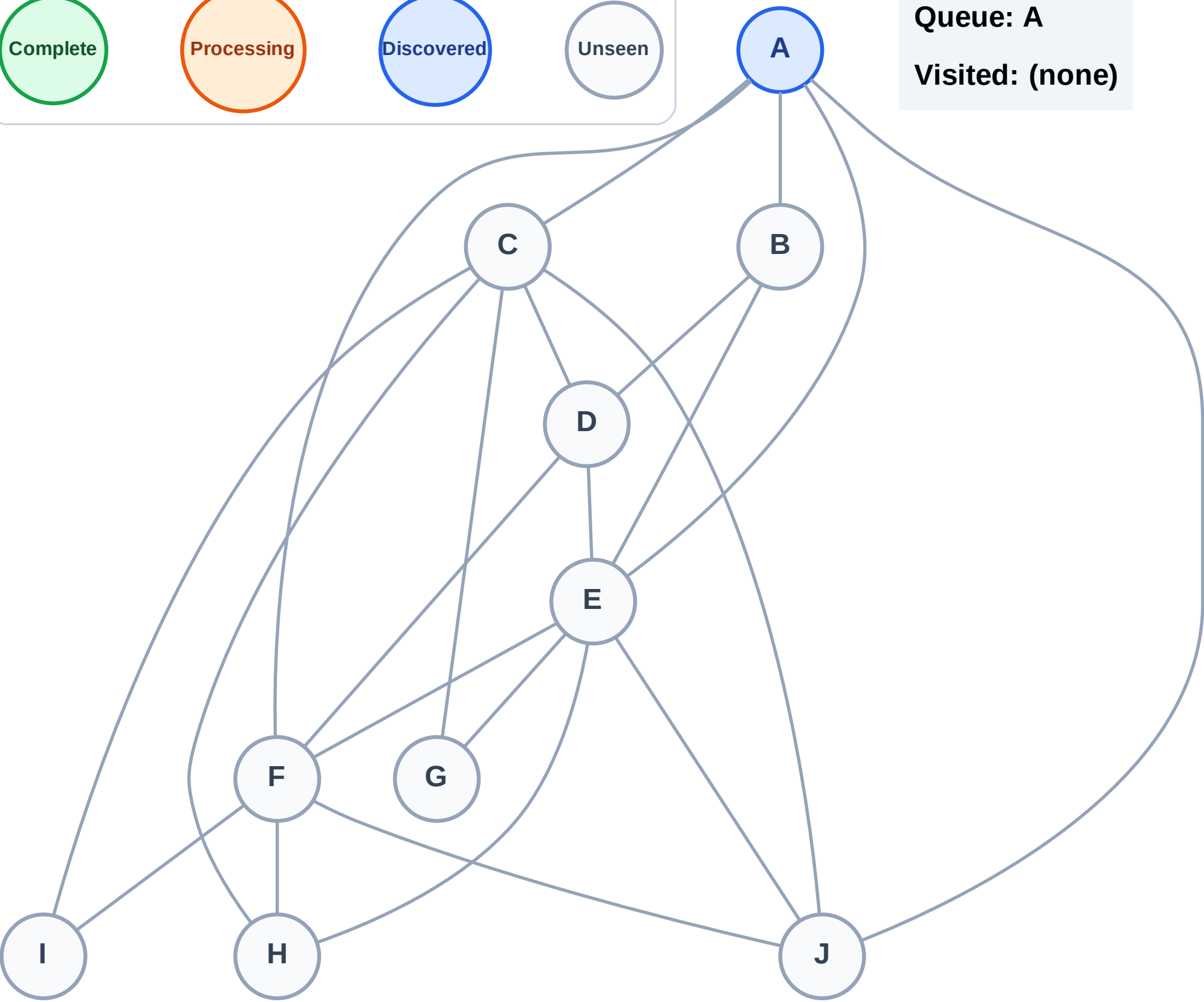
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

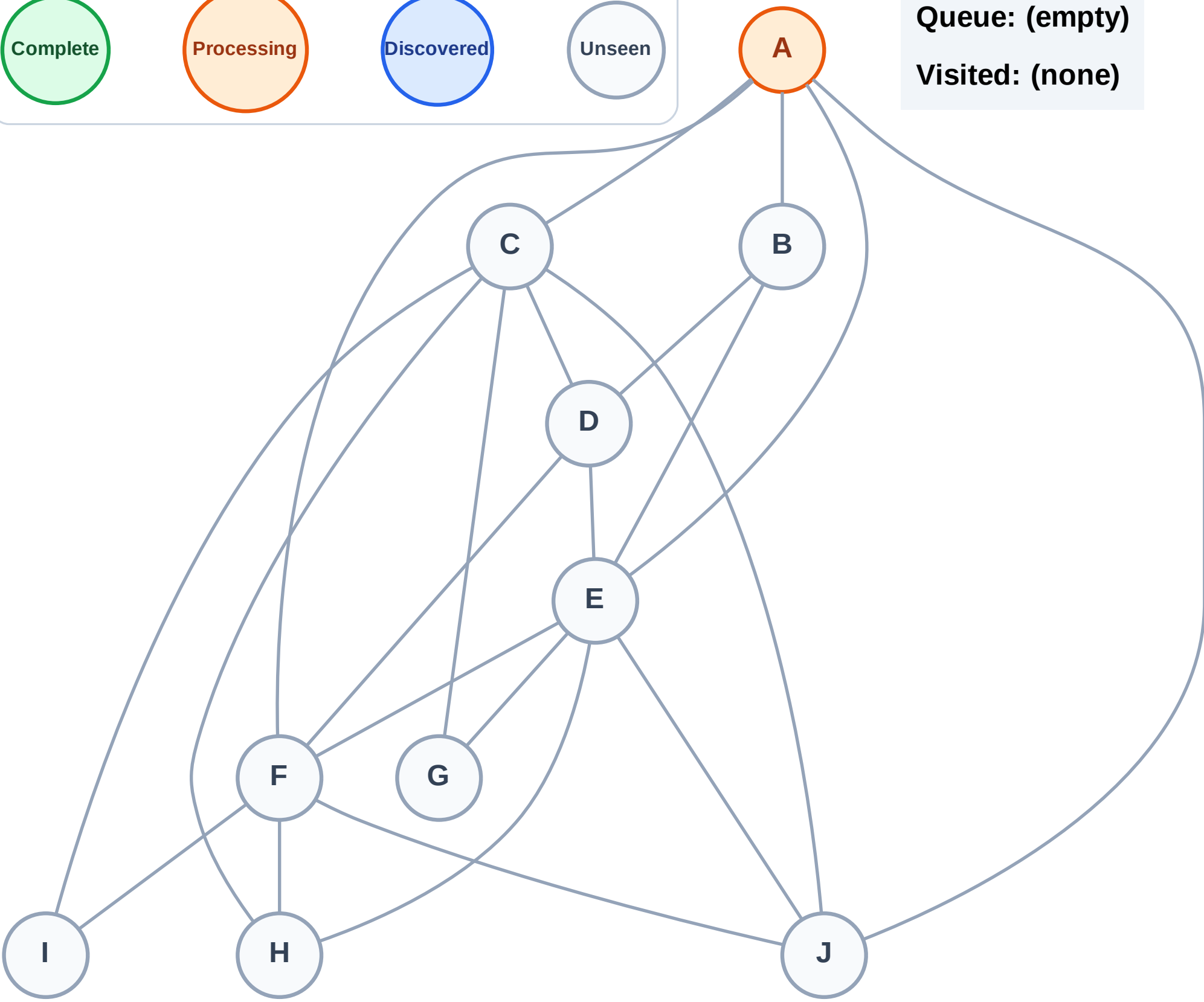
Step 2: Process node A

Legend



Queue: (empty)

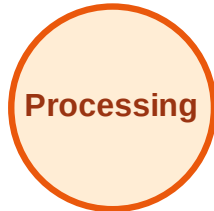
Visited: (none)



BFS Traversal

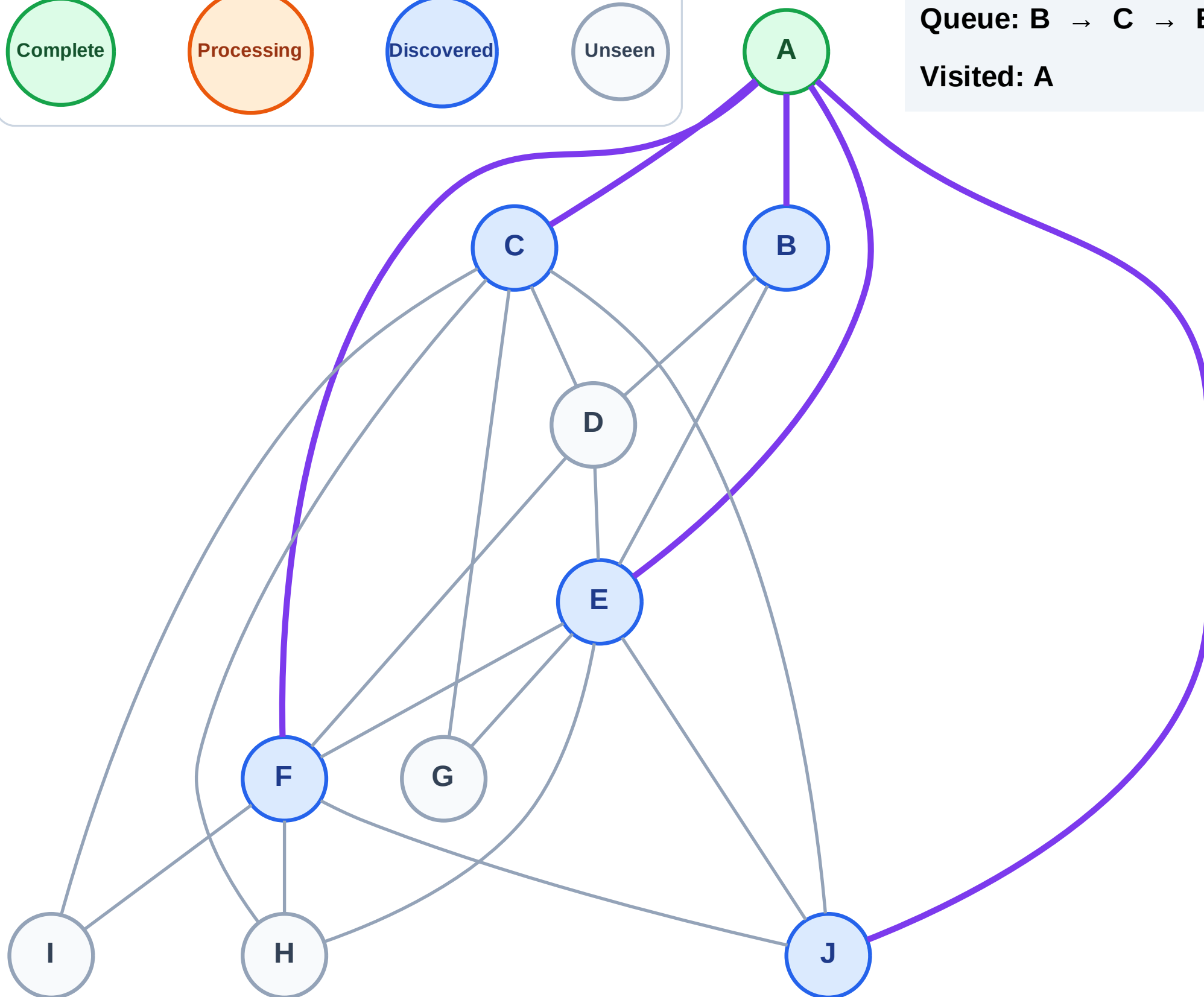
Step 3: Discover B, C, E, F, J from A

Legend



Queue: B → C → E → F → J

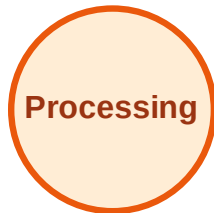
Visited: A



BFS Traversal

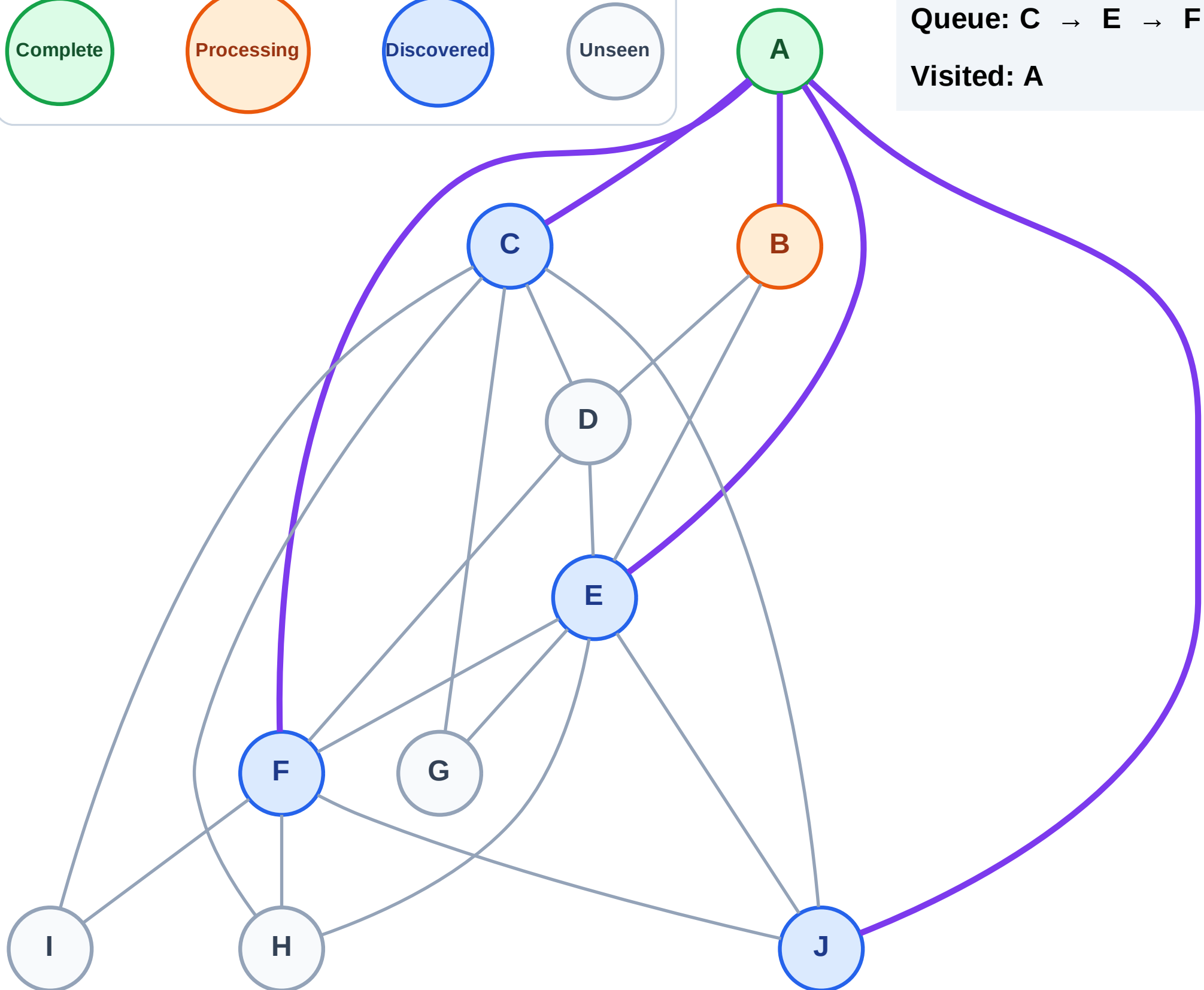
Step 4: Process node B

Legend



Queue: C → E → F → J

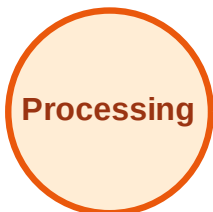
Visited: A



BFS Traversal

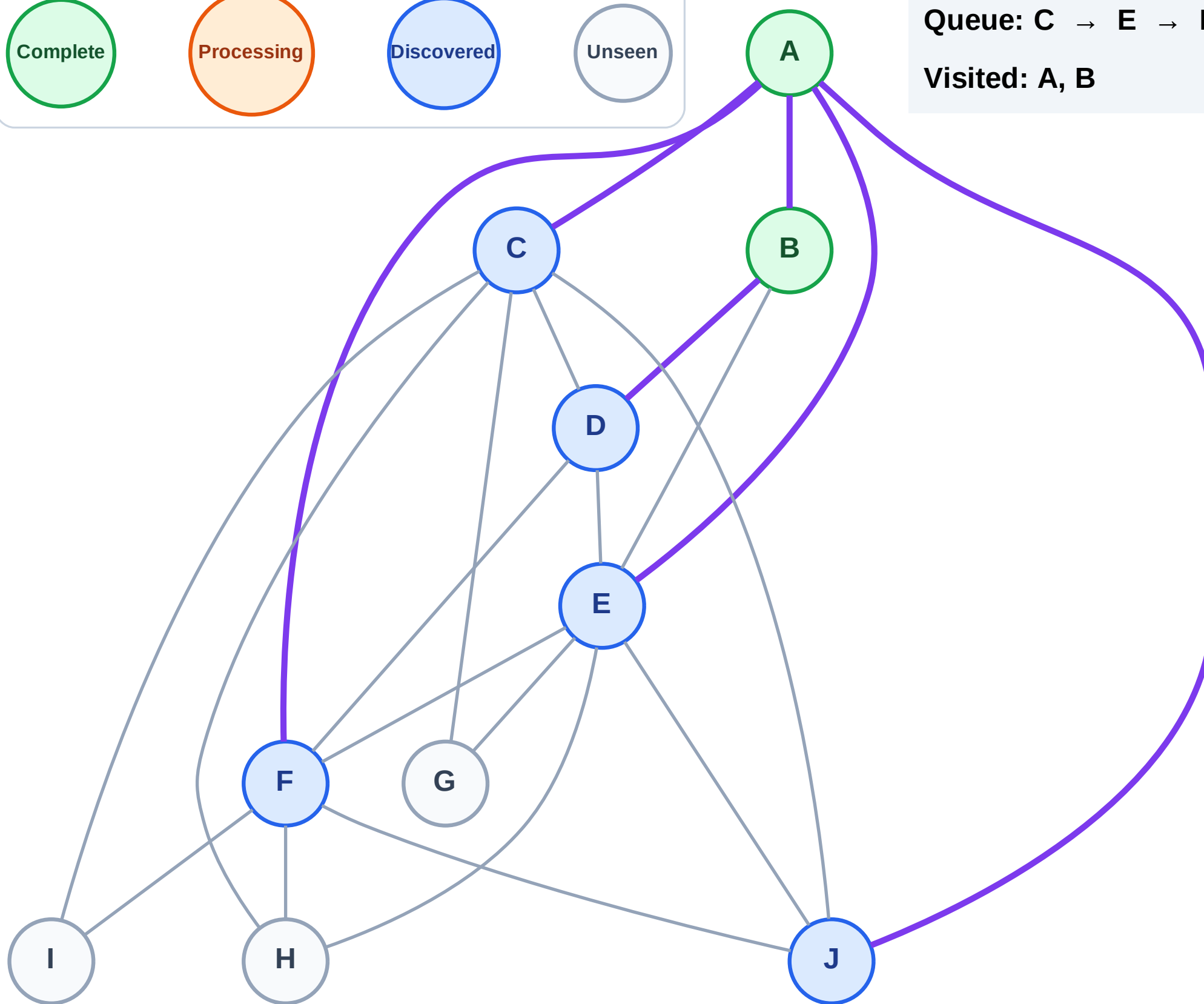
Step 5: Discover D from B

Legend



Queue: C → E → F → J → D

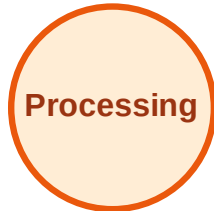
Visited: A, B



BFS Traversal

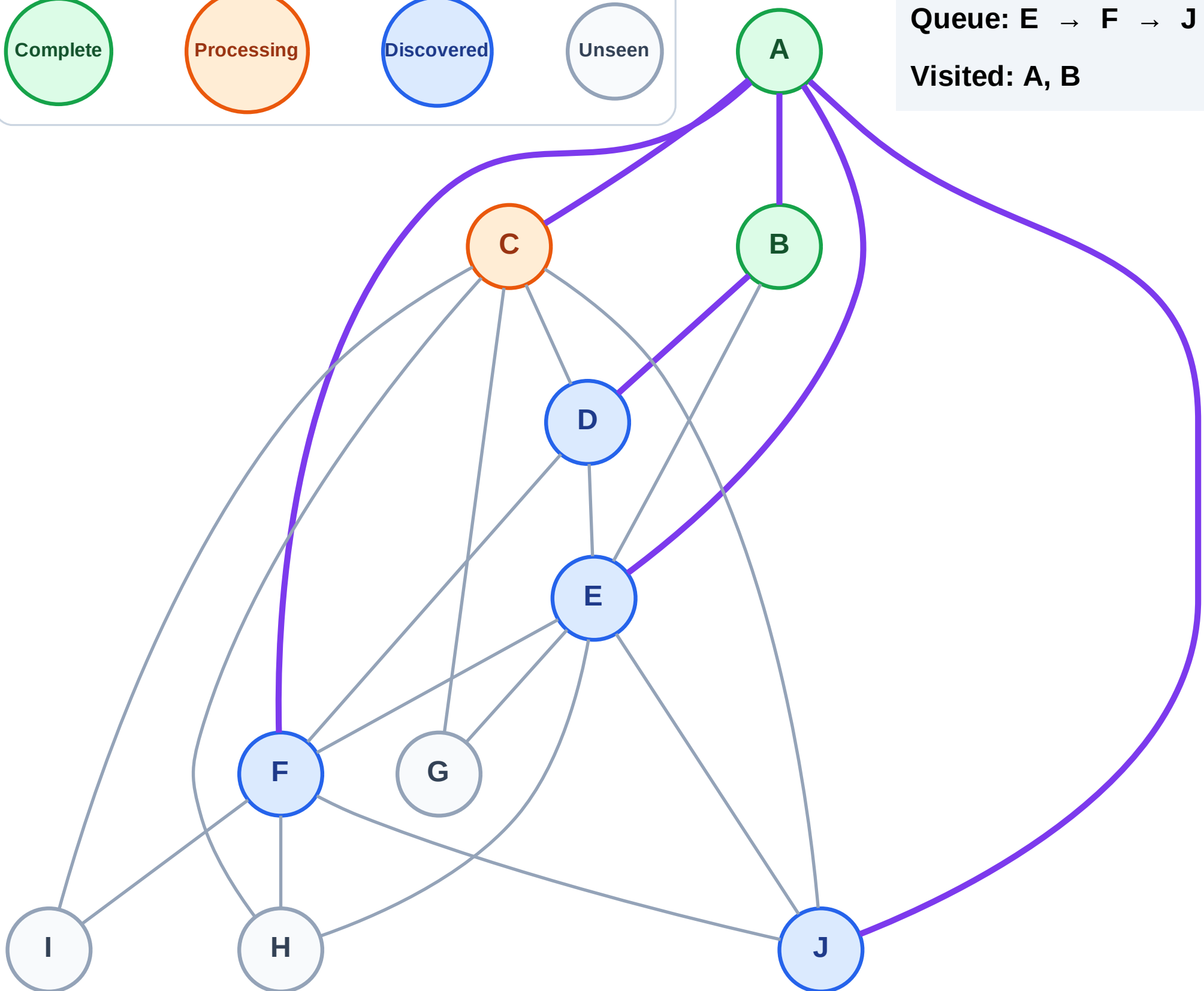
Step 6: Process node C

Legend



Queue: E → F → J → D

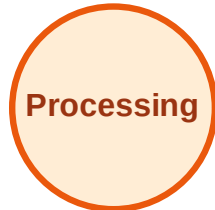
Visited: A, B



BFS Traversal

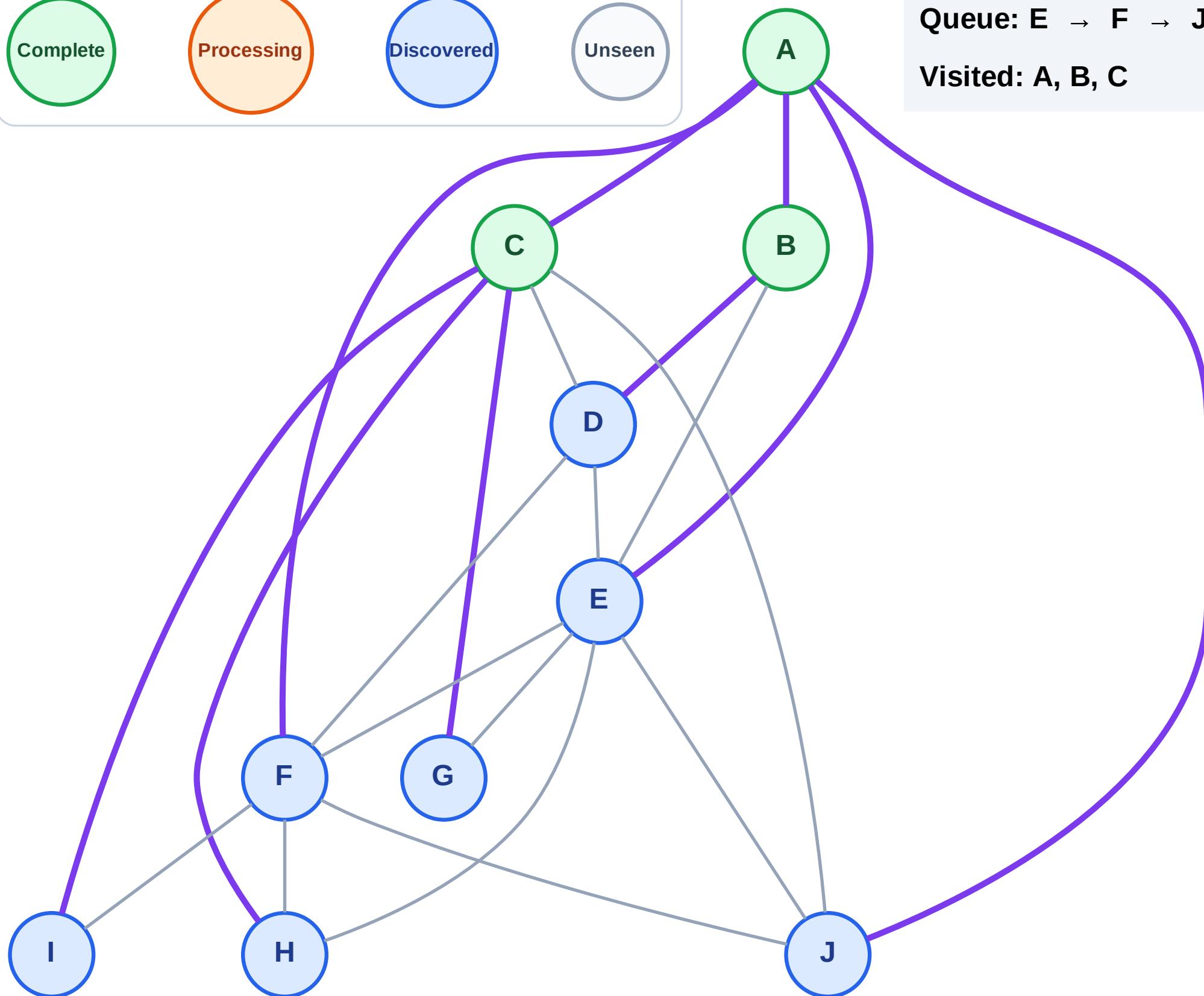
Step 7: Discover G, H, I from C

Legend



Queue: E → F → J → D → G → H → I

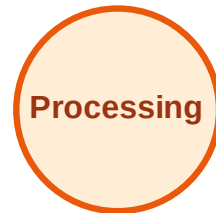
Visited: A, B, C



BFS Traversal

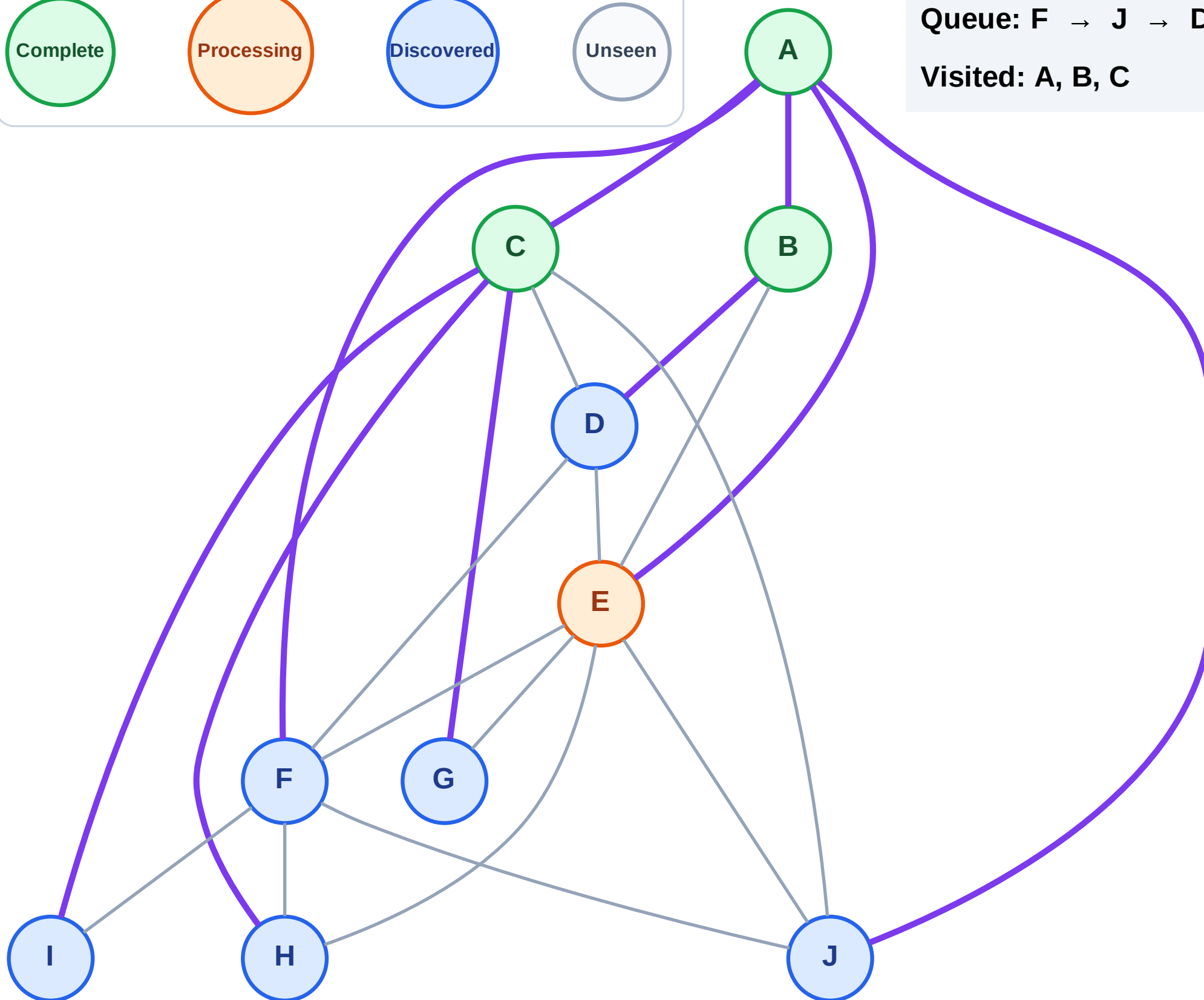
Step 8: Process node E

Legend



Queue: F → J → D → G → H → I

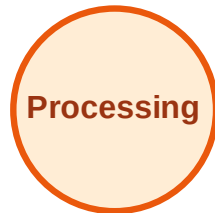
Visited: A, B, C



BFS Traversal

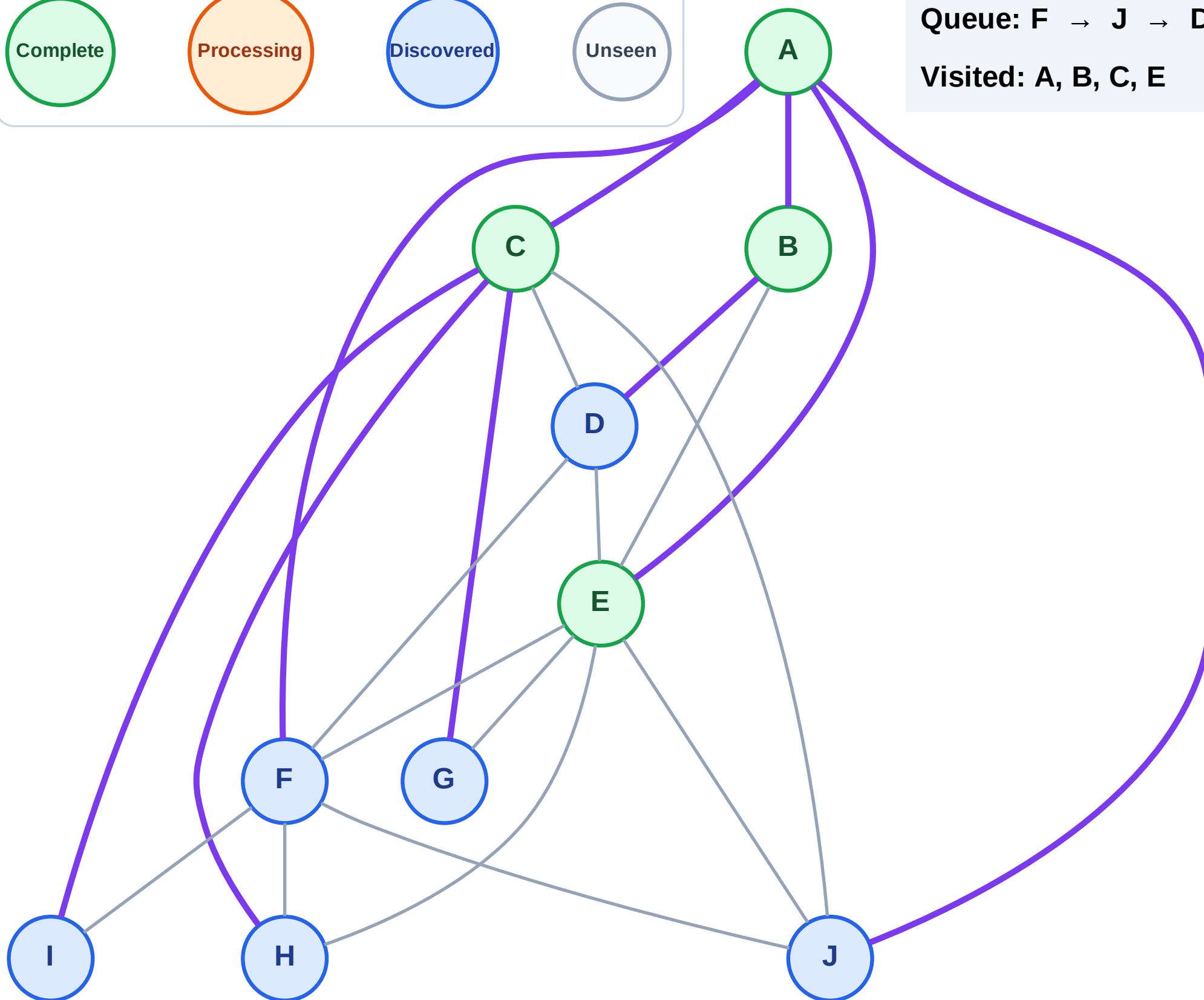
Step 9: No new neighbors from E

Legend



Queue: F → J → D → G → H → I

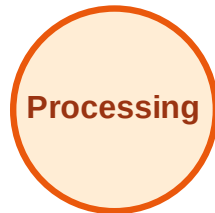
Visited: A, B, C, E



BFS Traversal

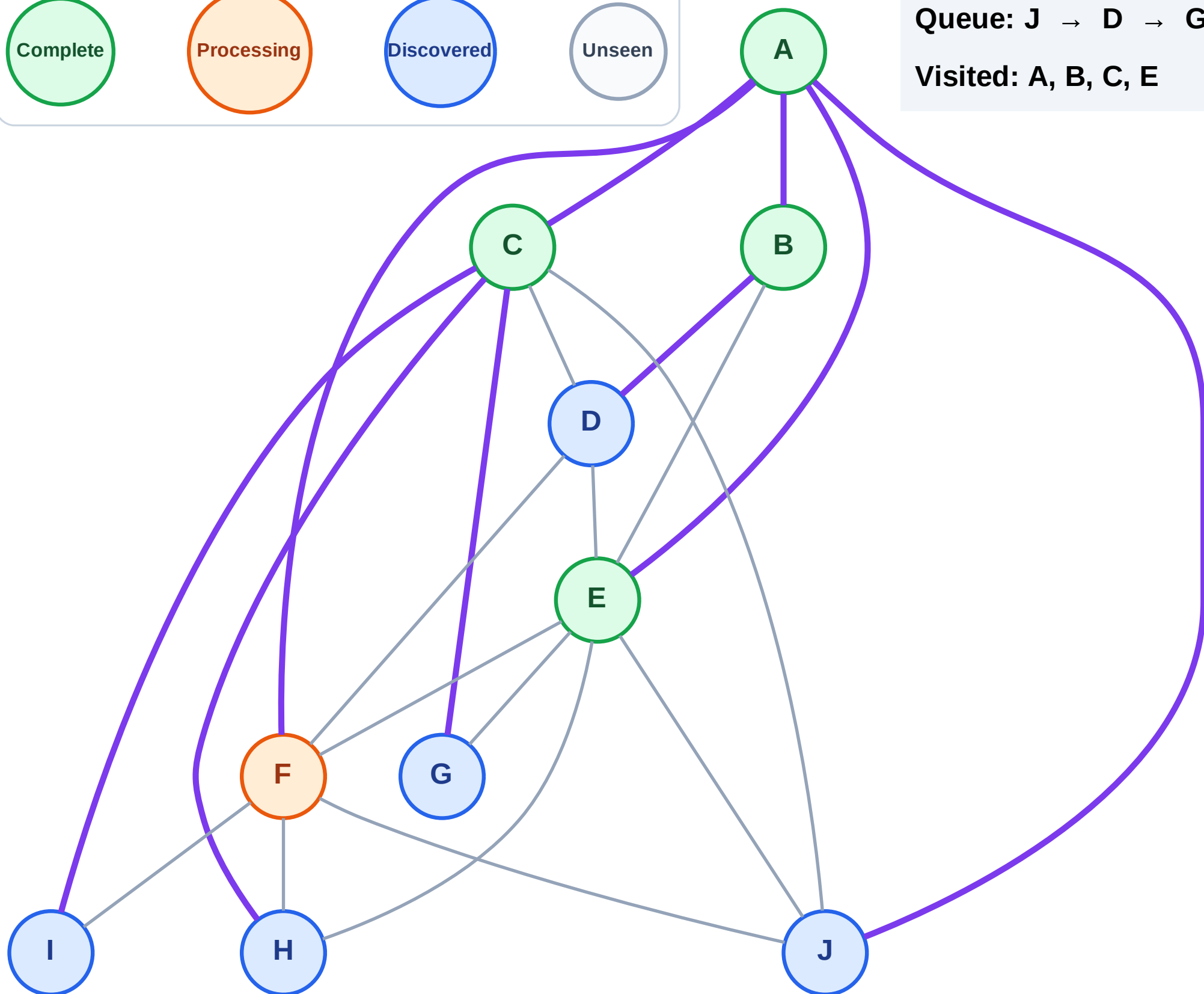
Step 10: Process node F

Legend



Queue: J → D → G → H → I

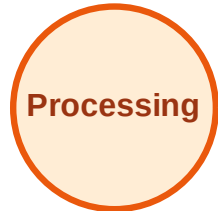
Visited: A, B, C, E



BFS Traversal

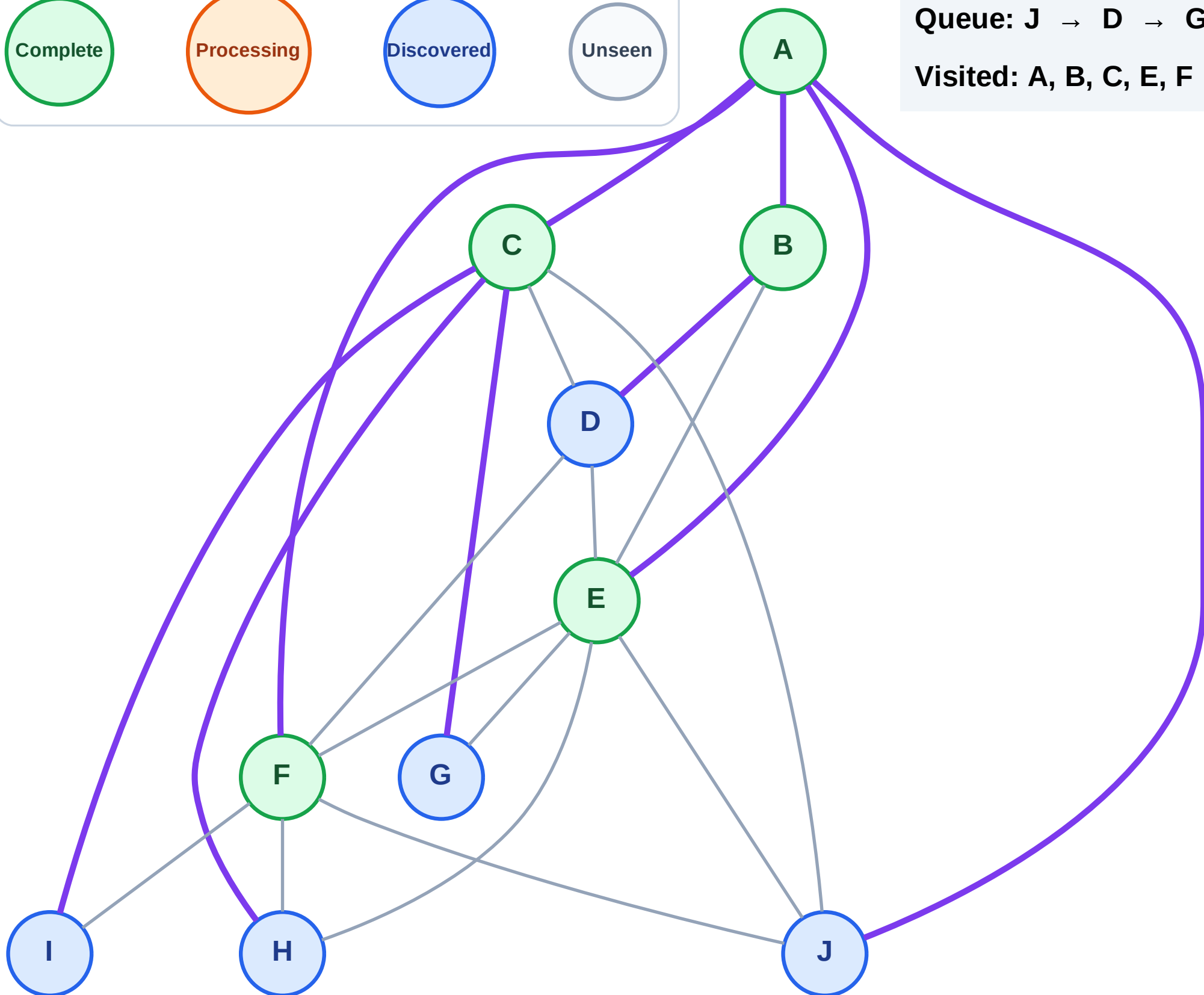
Step 11: No new neighbors from F

Legend



Queue: J → D → G → H → I

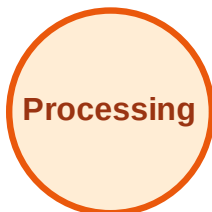
Visited: A, B, C, E, F



BFS Traversal

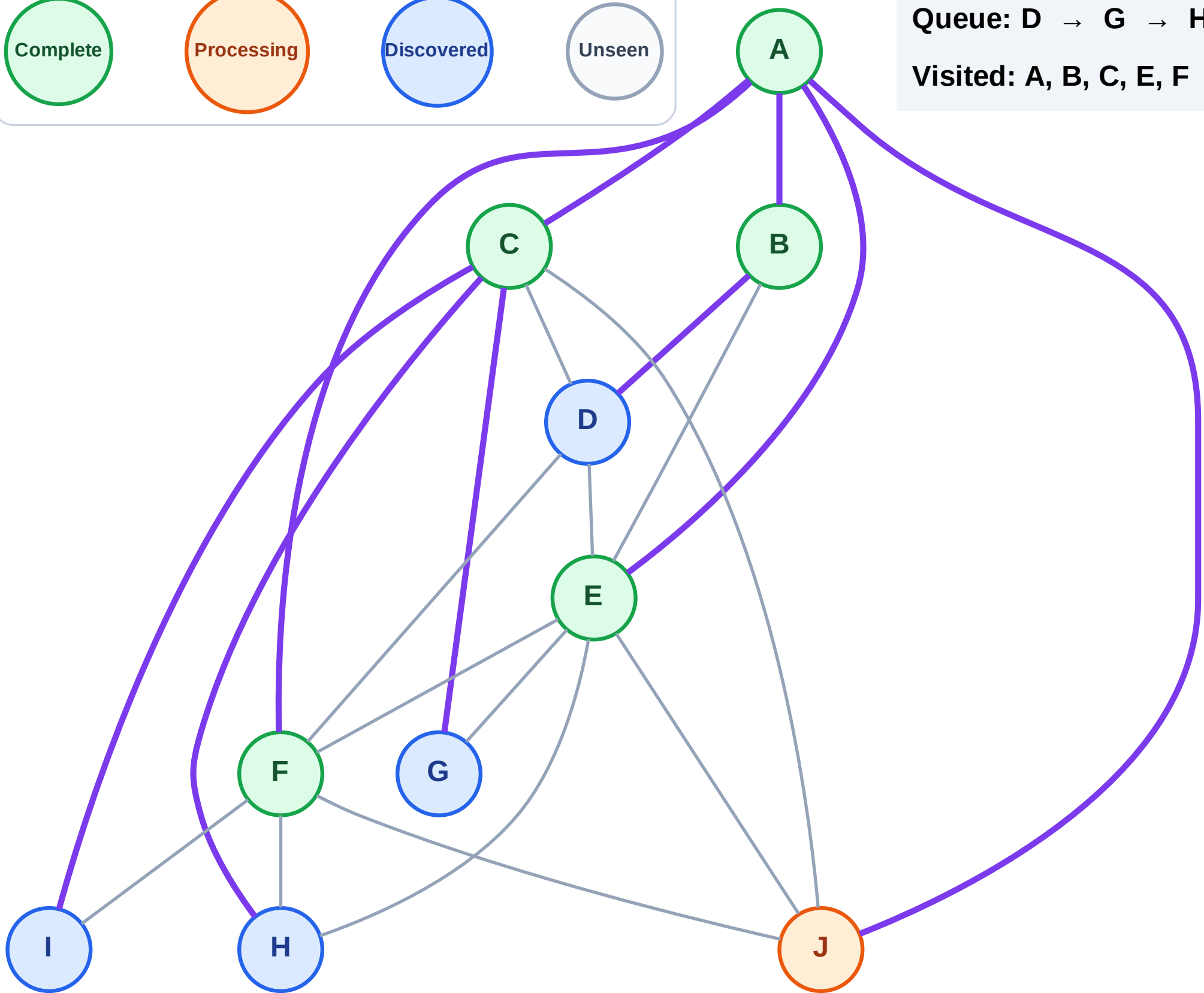
Step 12: Process node J

Legend



Queue: D → G → H → I

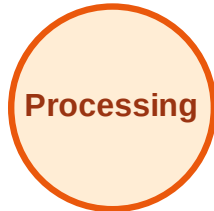
Visited: A, B, C, E, F



BFS Traversal

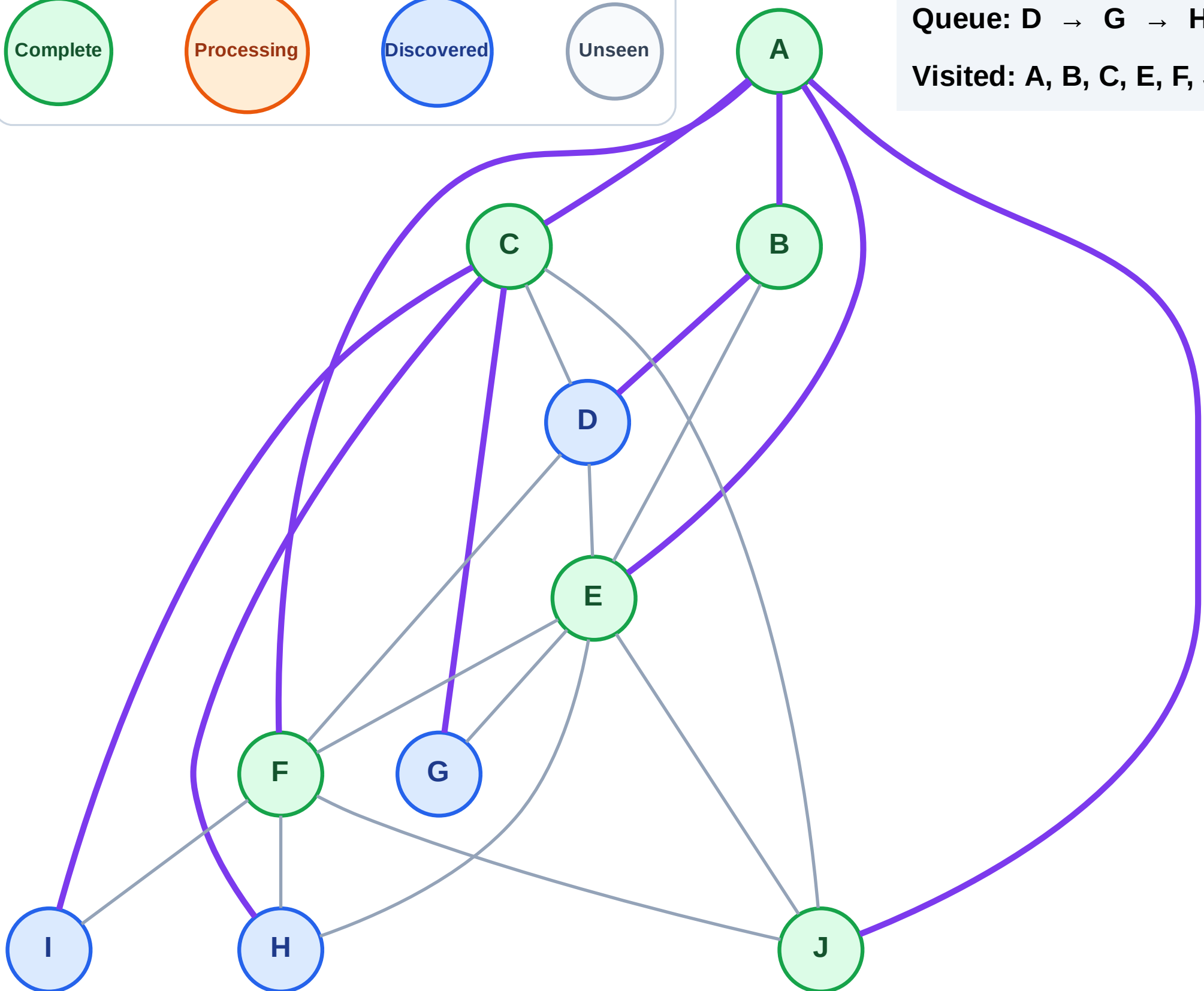
Step 13: No new neighbors from J

Legend



Queue: D → G → H → I

Visited: A, B, C, E, F, J



BFS Traversal

Step 14: Process node D

Legend

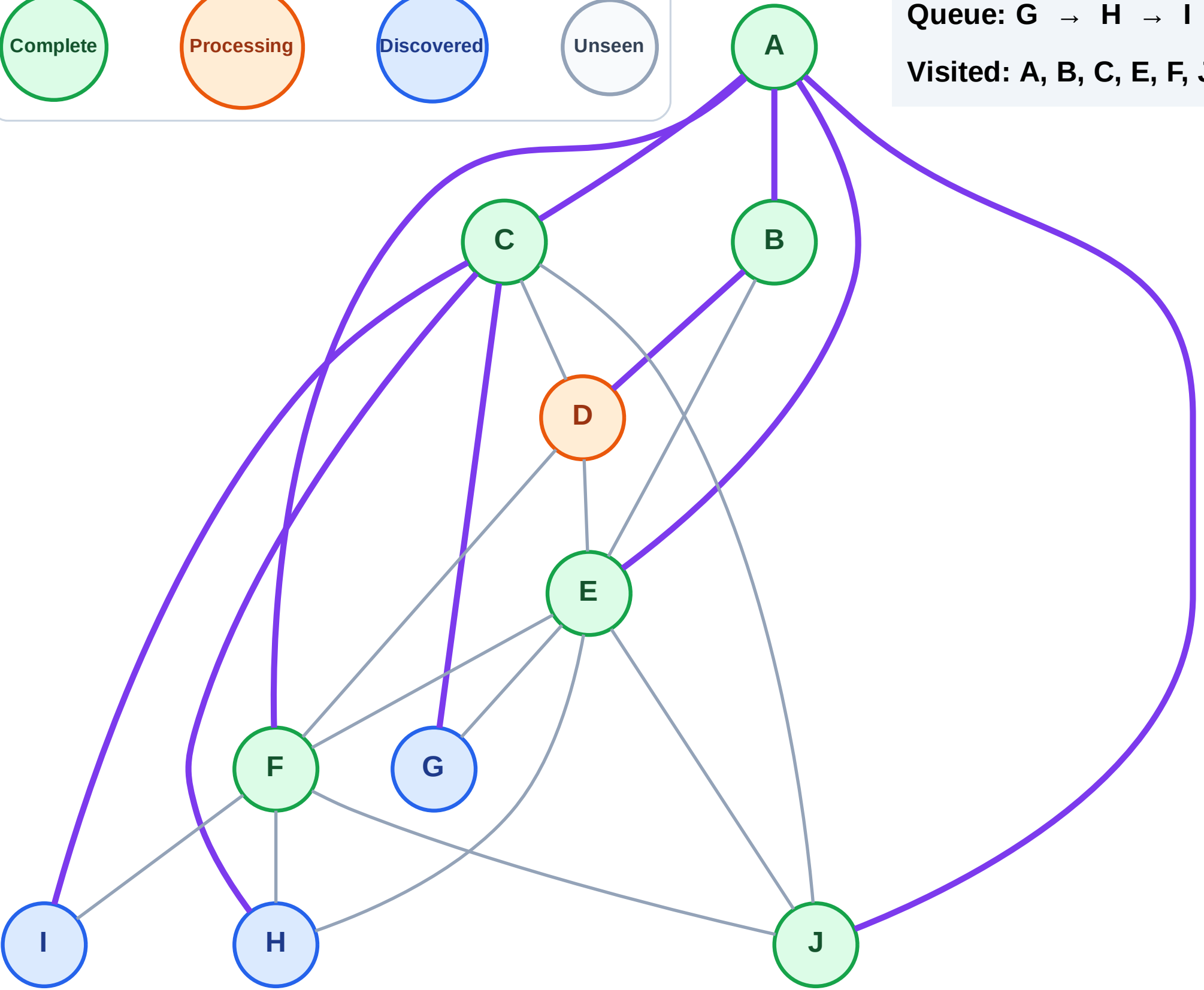
Complete

Processing

Discovered

Unseen

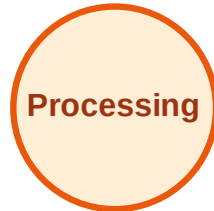
Queue: G → H → I
Visited: A, B, C, E, F, J



BFS Traversal

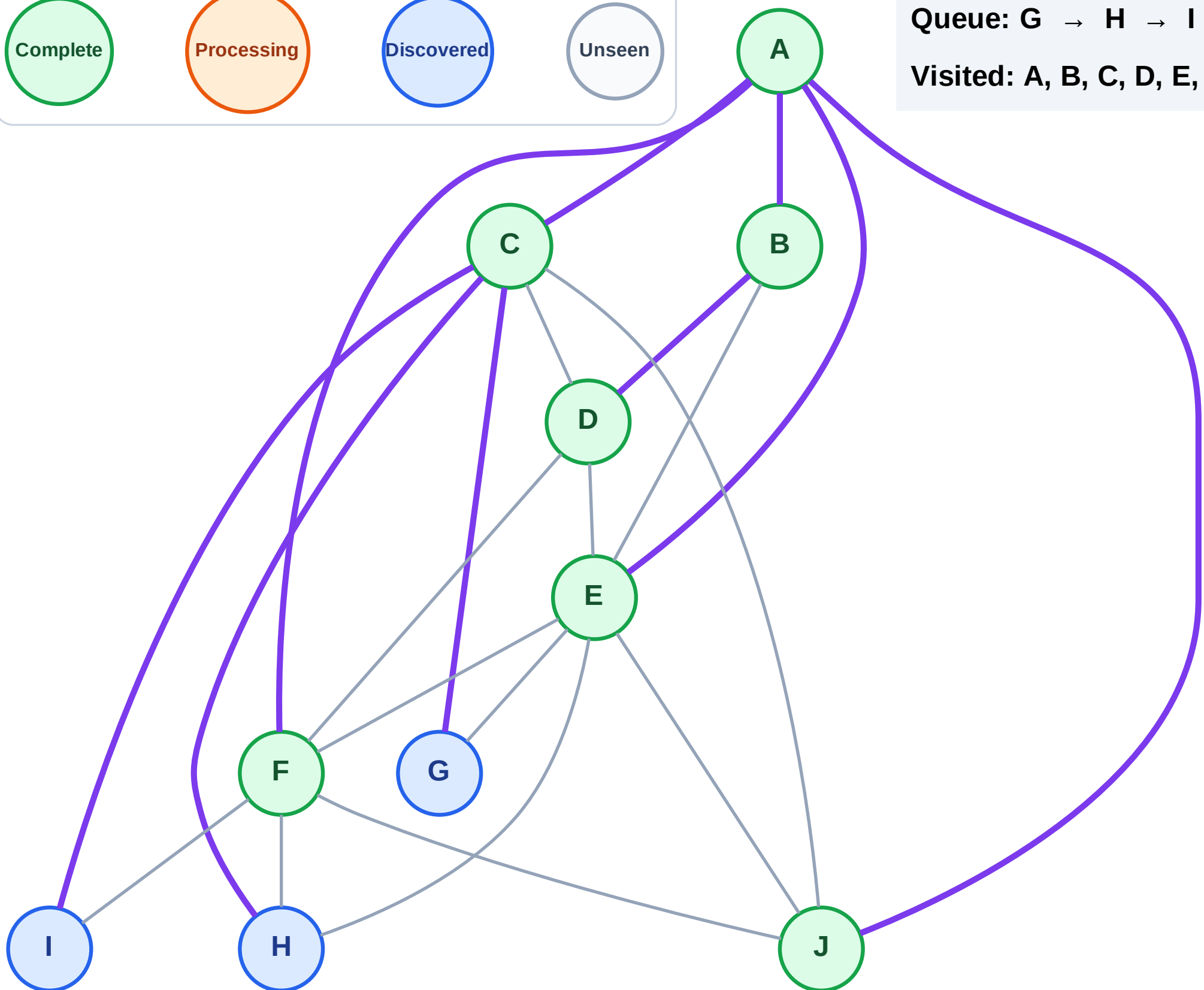
Step 15: No new neighbors from D

Legend



Queue: G → H → I

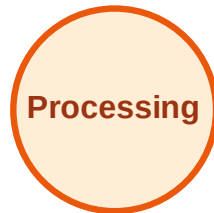
Visited: A, B, C, D, E, F, J



BFS Traversal

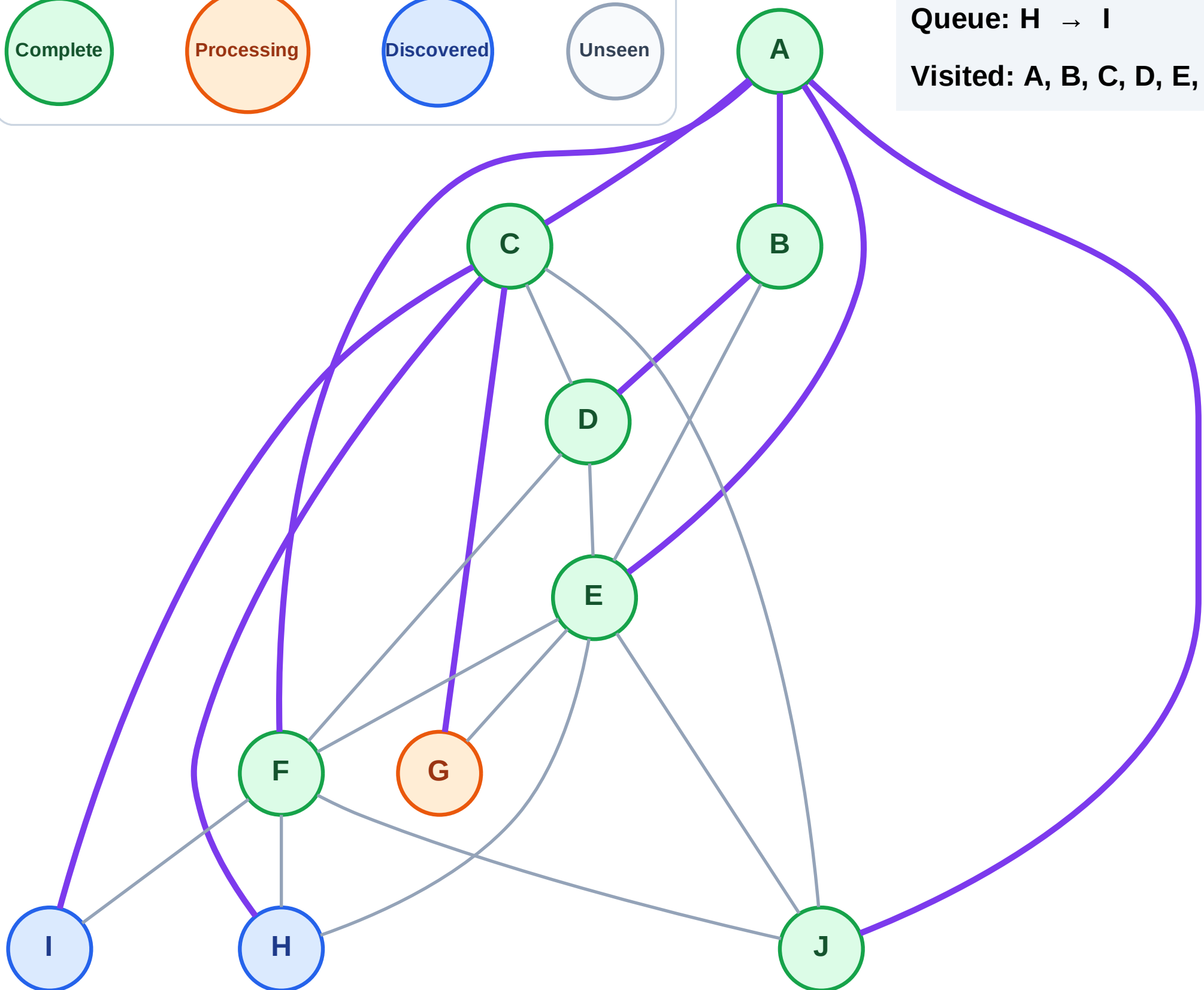
Step 16: Process node G

Legend



Queue: H → I

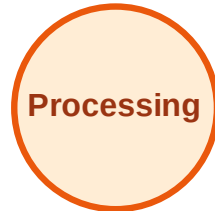
Visited: A, B, C, D, E, F, J



BFS Traversal

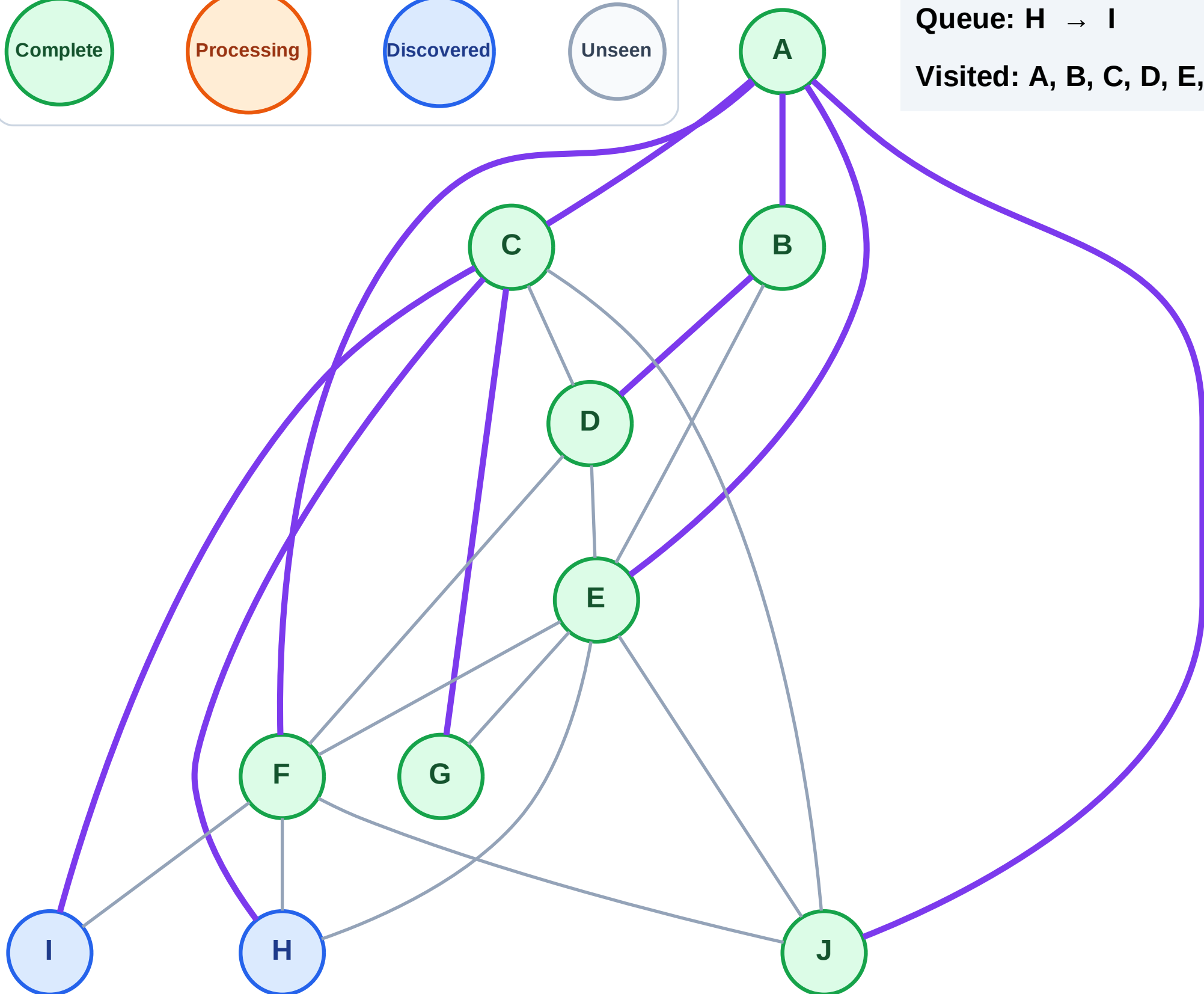
Step 17: No new neighbors from G

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node H

Legend

Complete

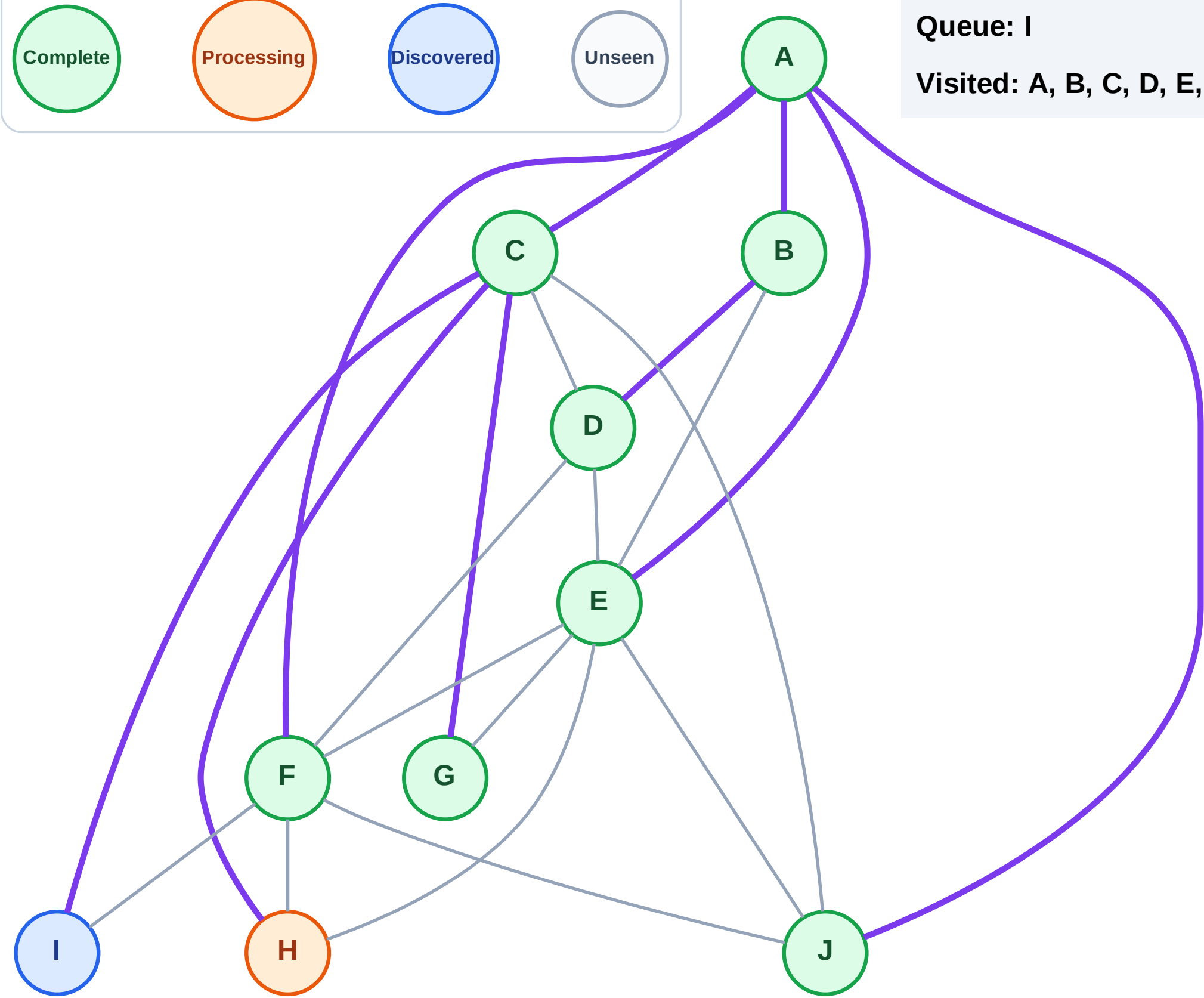
Processing

Discovered

Unseen

Queue: I

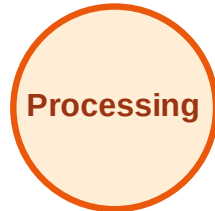
Visited: A, B, C, D, E, F, G, J



BFS Traversal

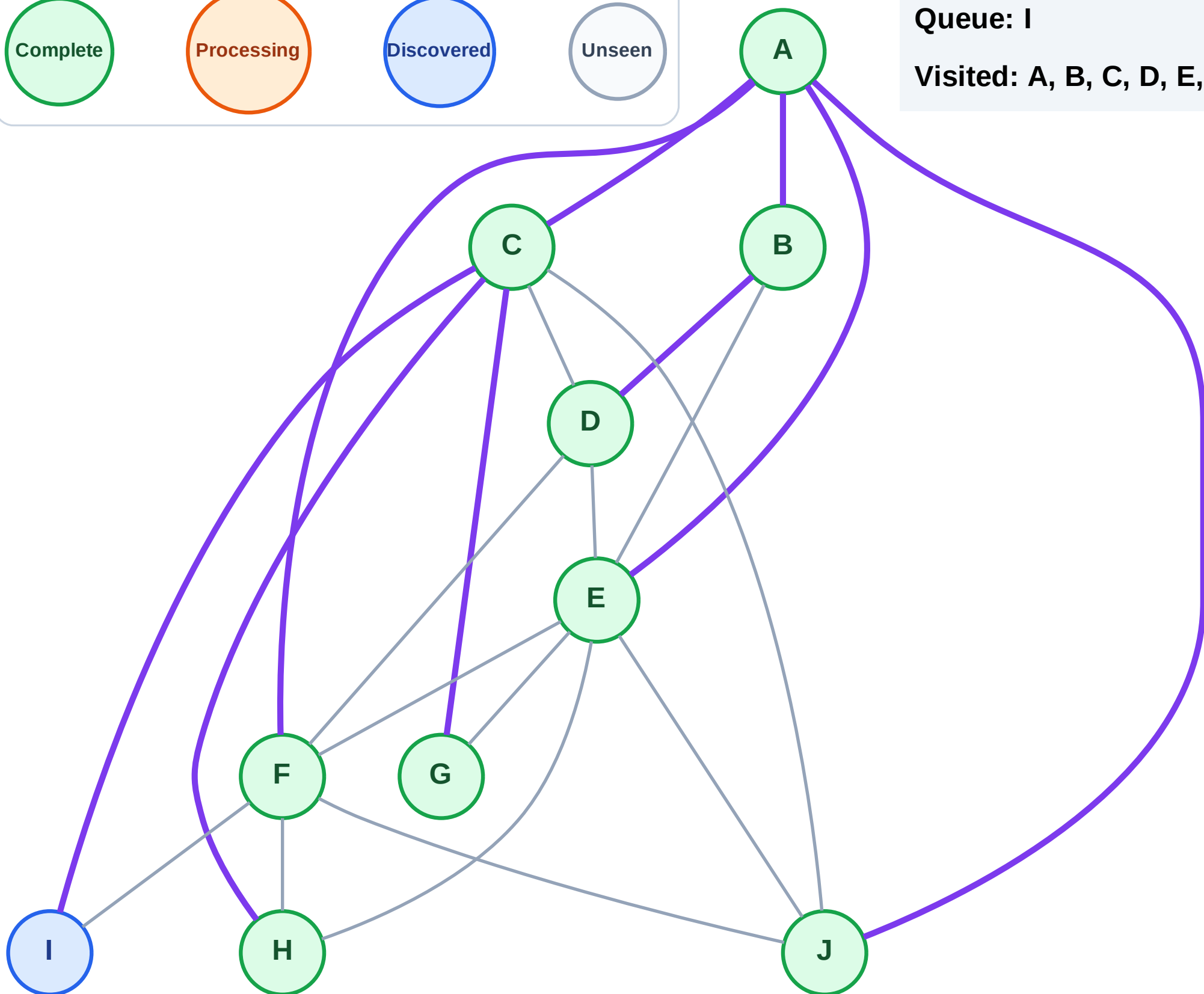
Step 19: No new neighbors from H

Legend



Queue: I

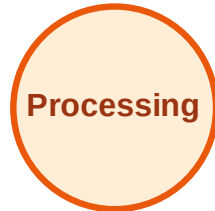
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

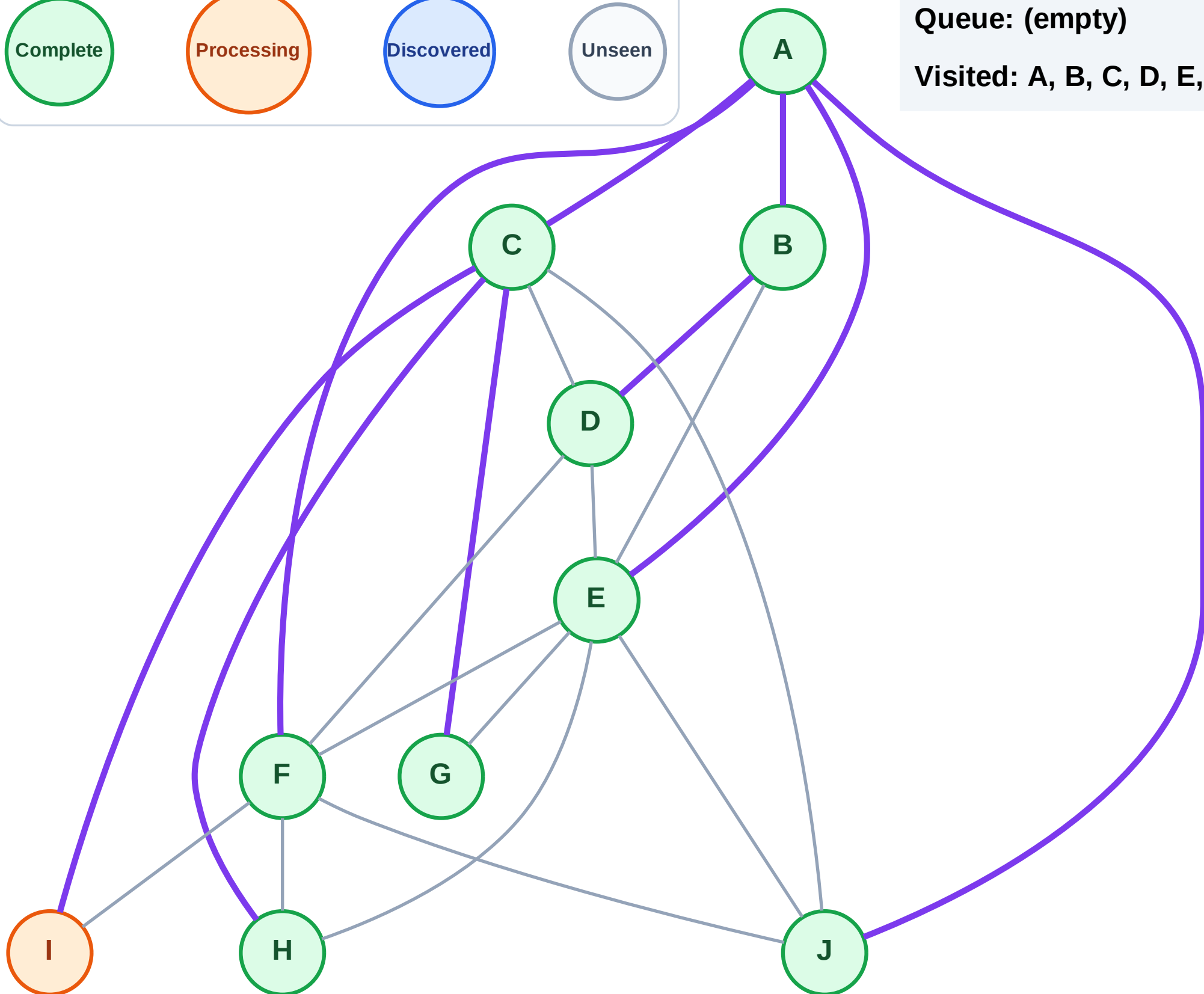
Step 20: Process node I

Legend



Queue: (empty)

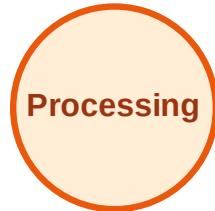
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

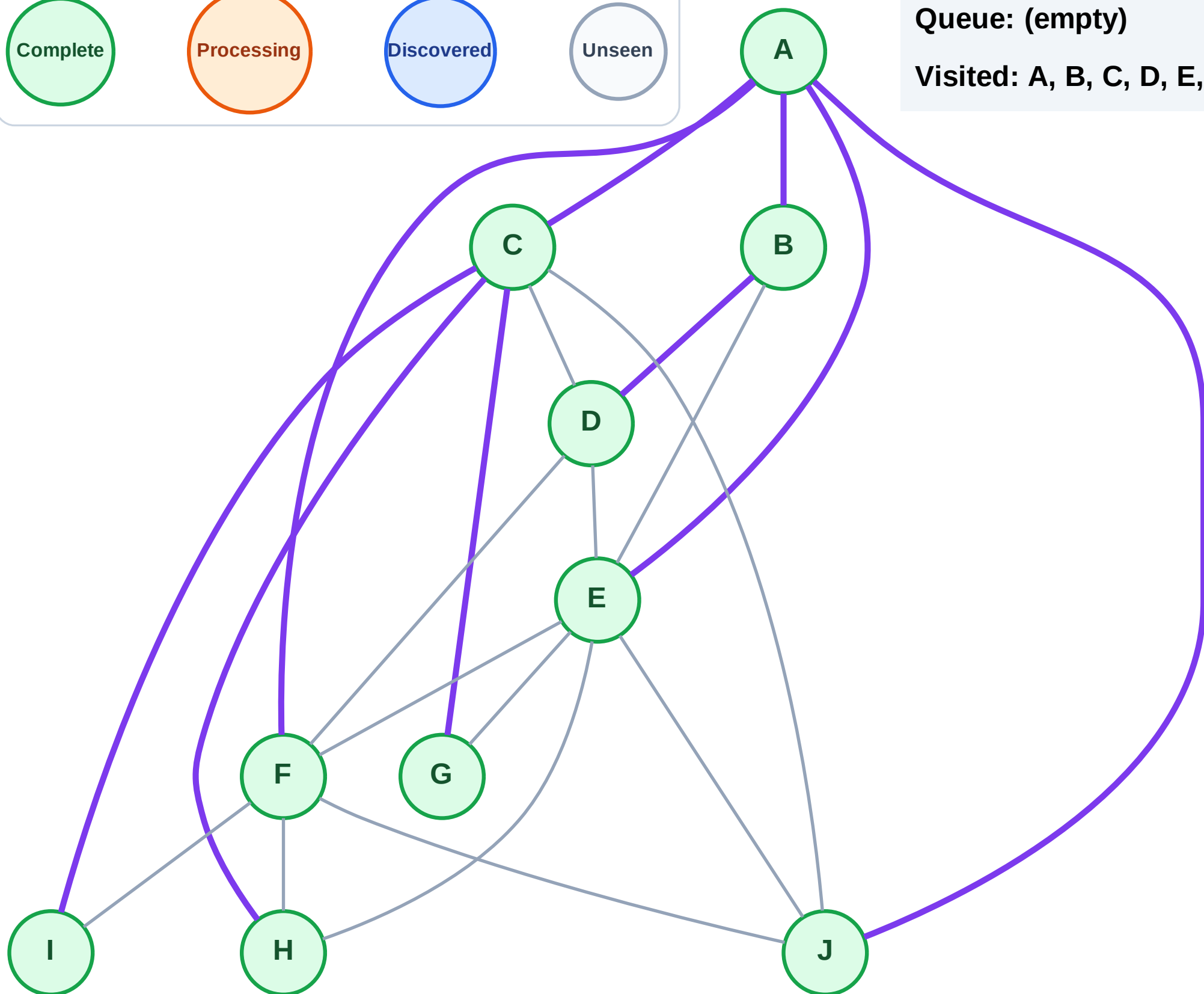
Step 21: No new neighbors from I

Legend



Queue: (empty)

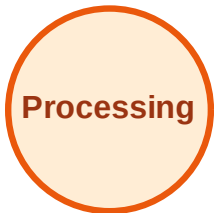
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

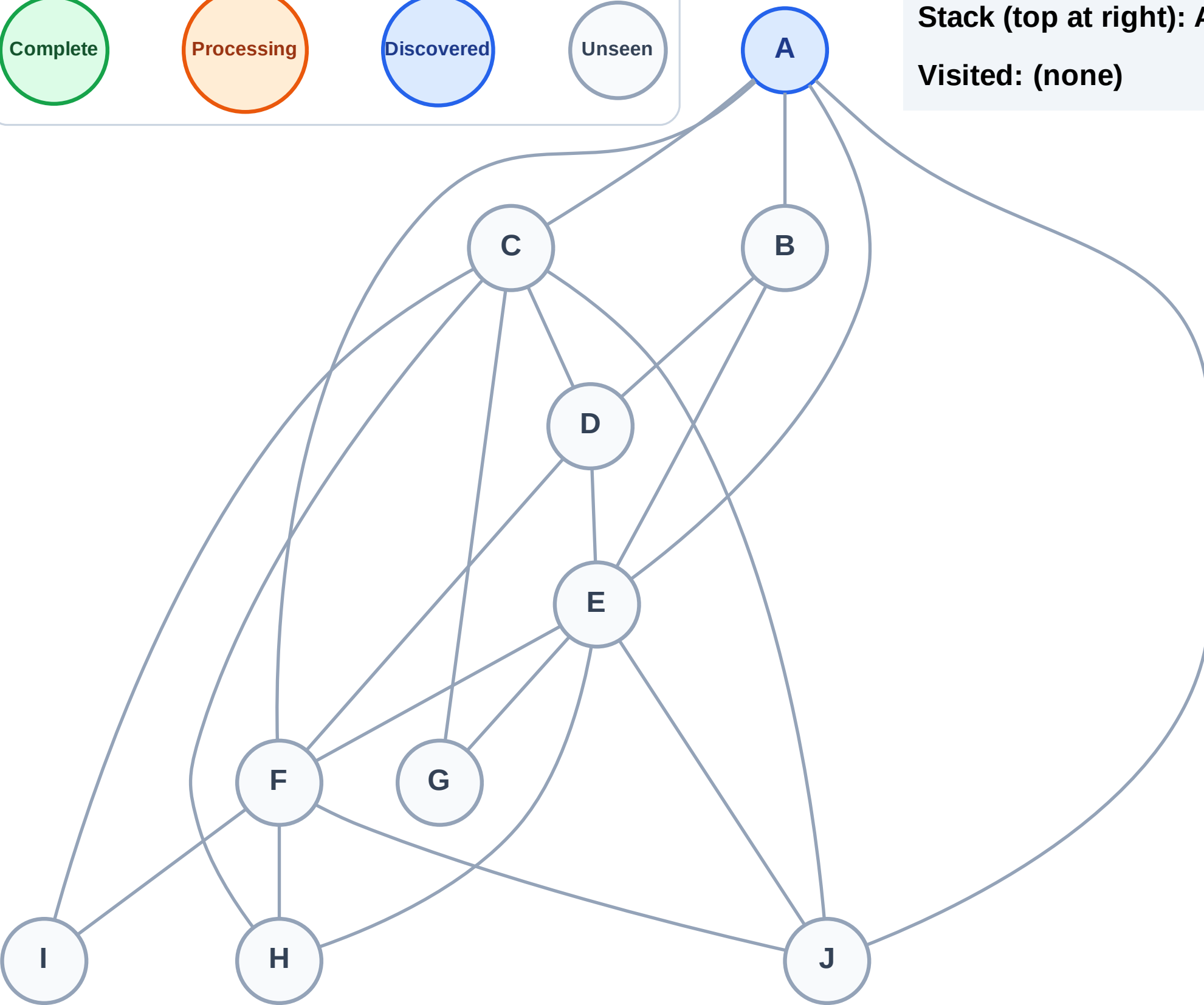
Step 1: Start at A

Legend



Stack (top at right): A

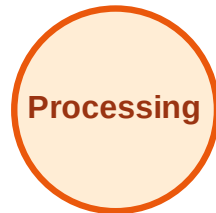
Visited: (none)



DFS Traversal

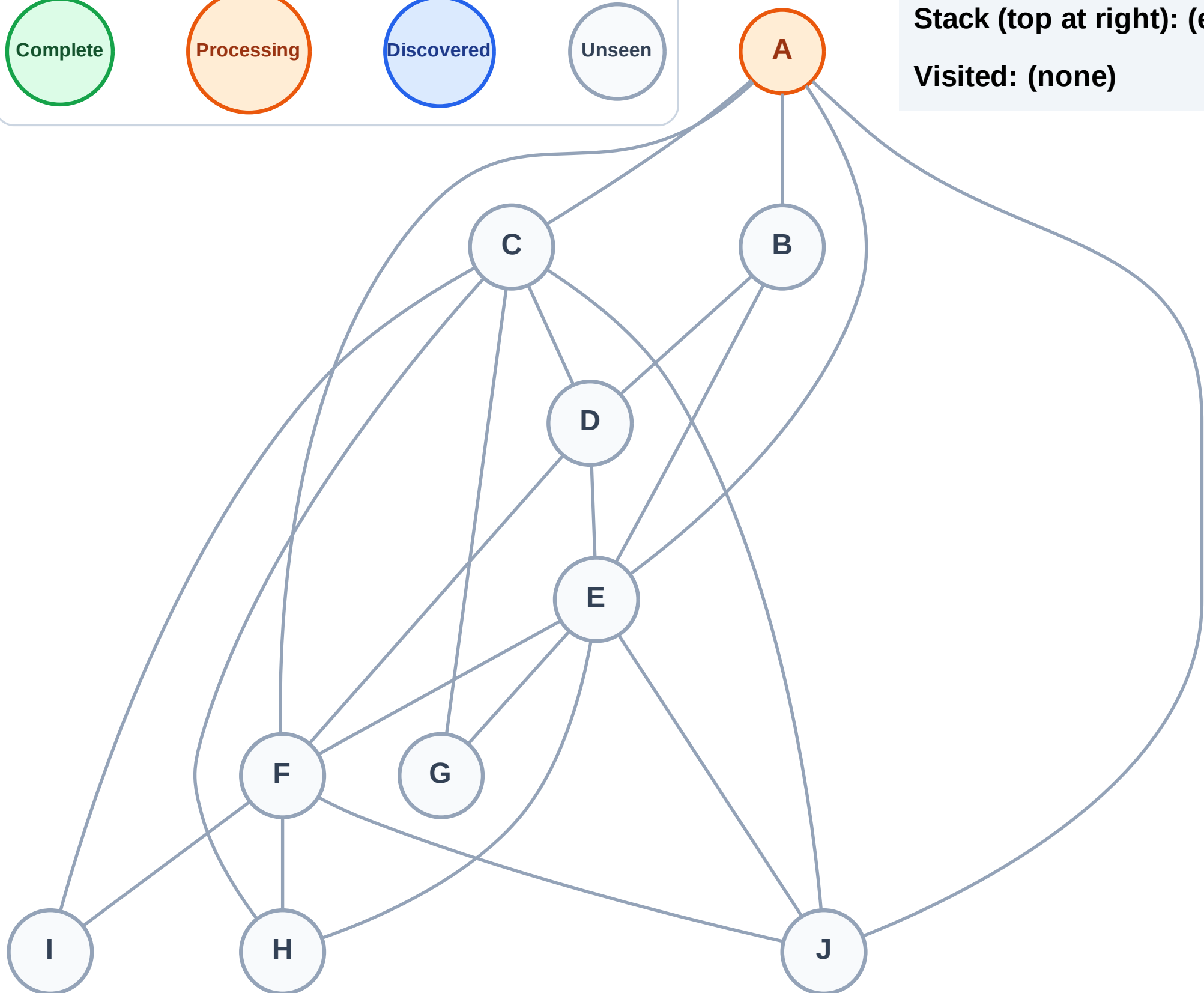
Step 2: Process node A

Legend



Stack (top at right): (empty)

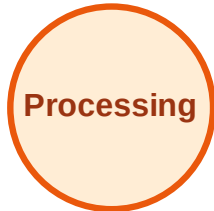
Visited: (none)



DFS Traversal

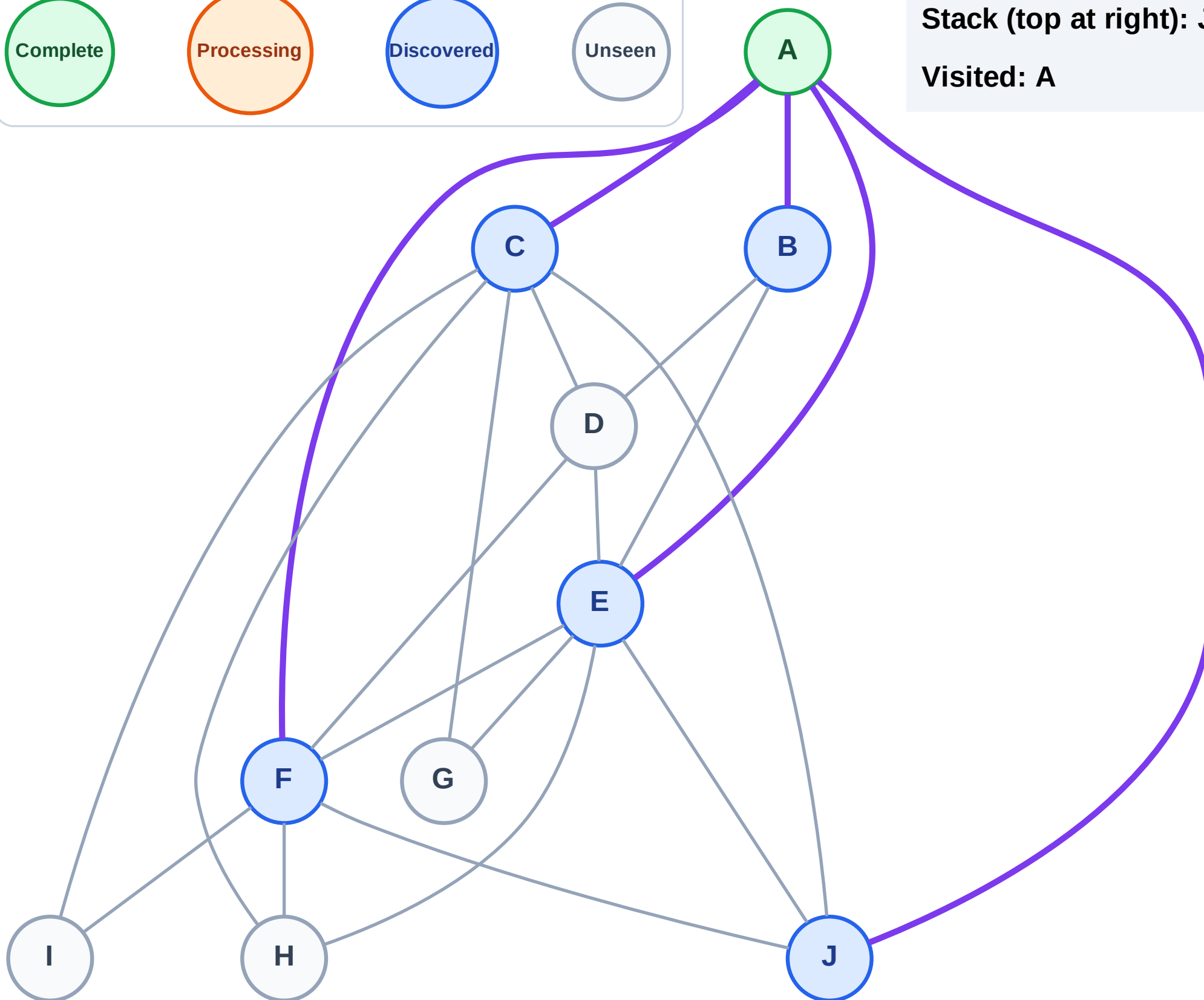
Step 3: Discover J, F, E, C, B from A

Legend



Stack (top at right): J | F | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

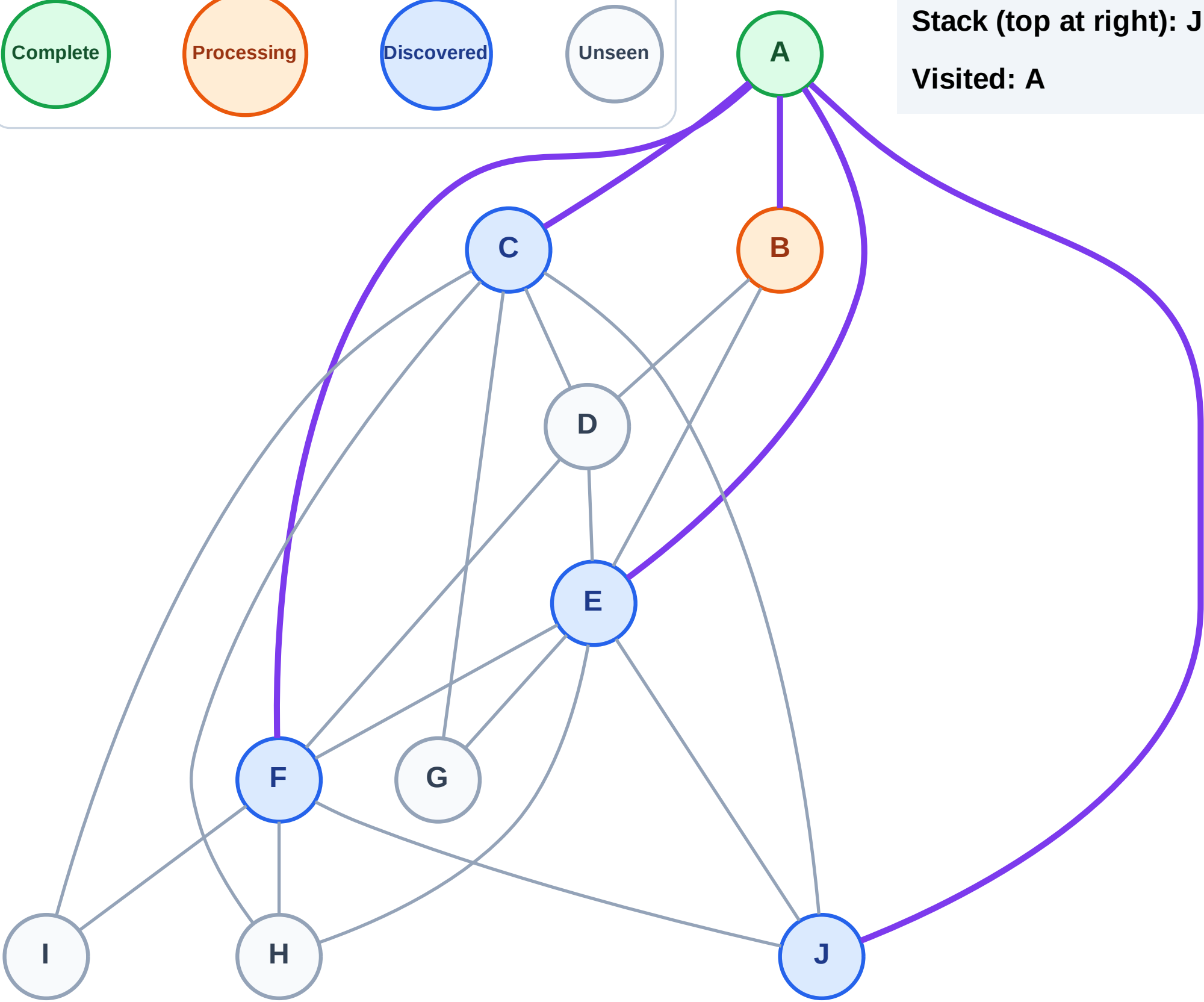
Complete

Processing

Discovered

Unseen

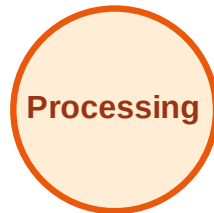
Stack (top at right): J | F | E | C
Visited: A



DFS Traversal

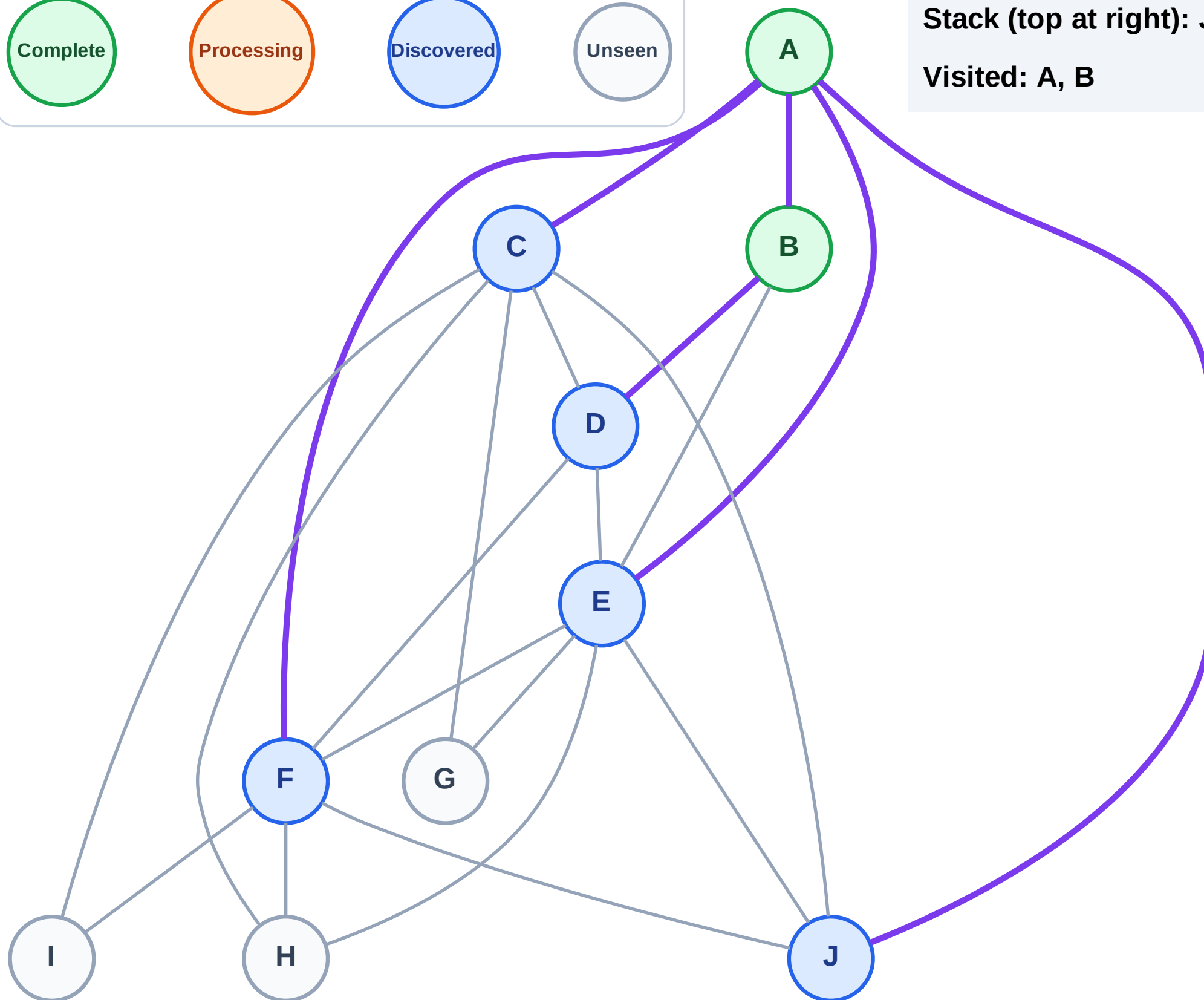
Step 5: Discover D from B

Legend



Stack (top at right): J | F | E | C | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

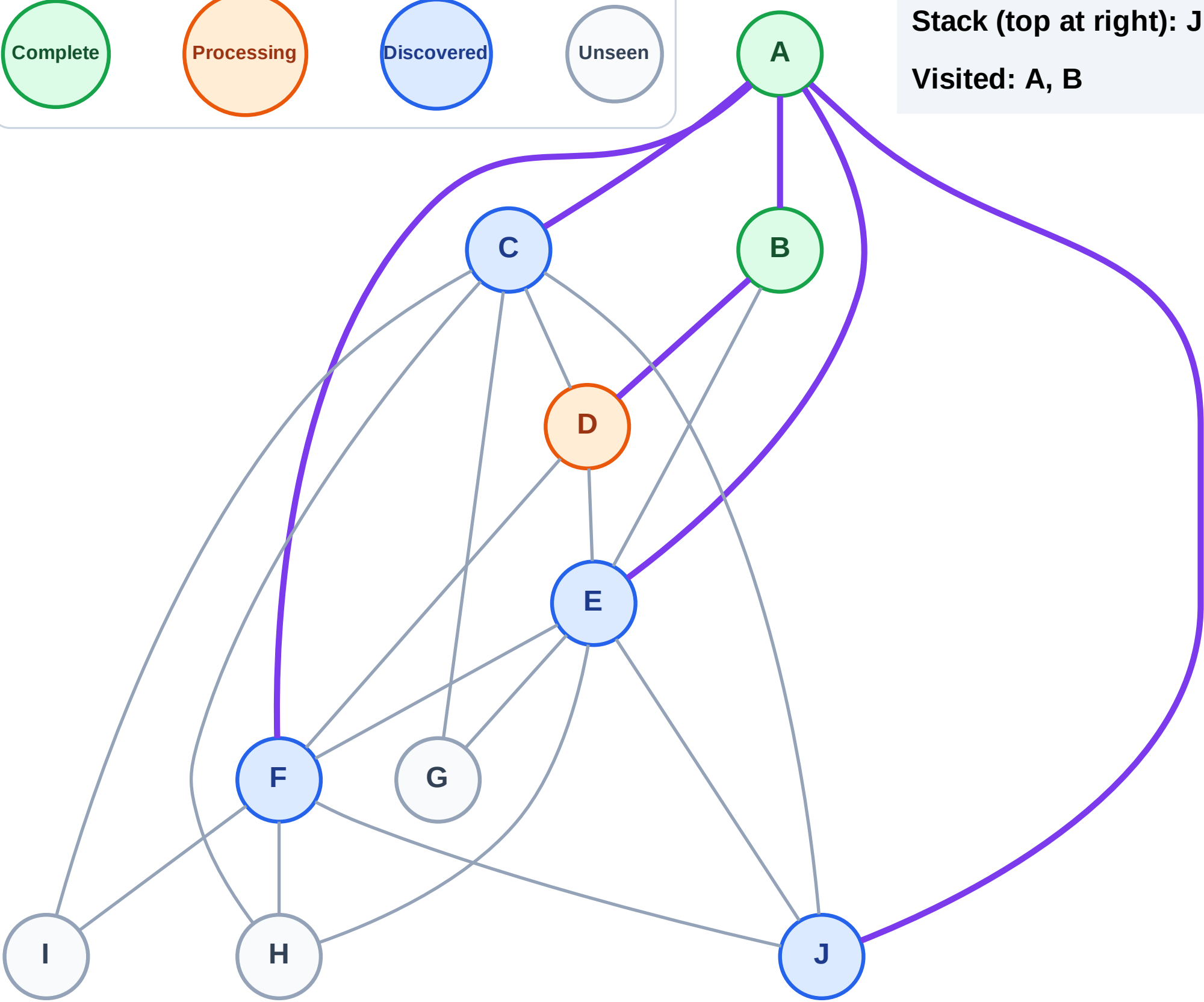
Complete

Processing

Discovered

Unseen

Stack (top at right): J | F | E | C
Visited: A, B



DFS Traversal

Step 7: No new neighbors from D

Legend

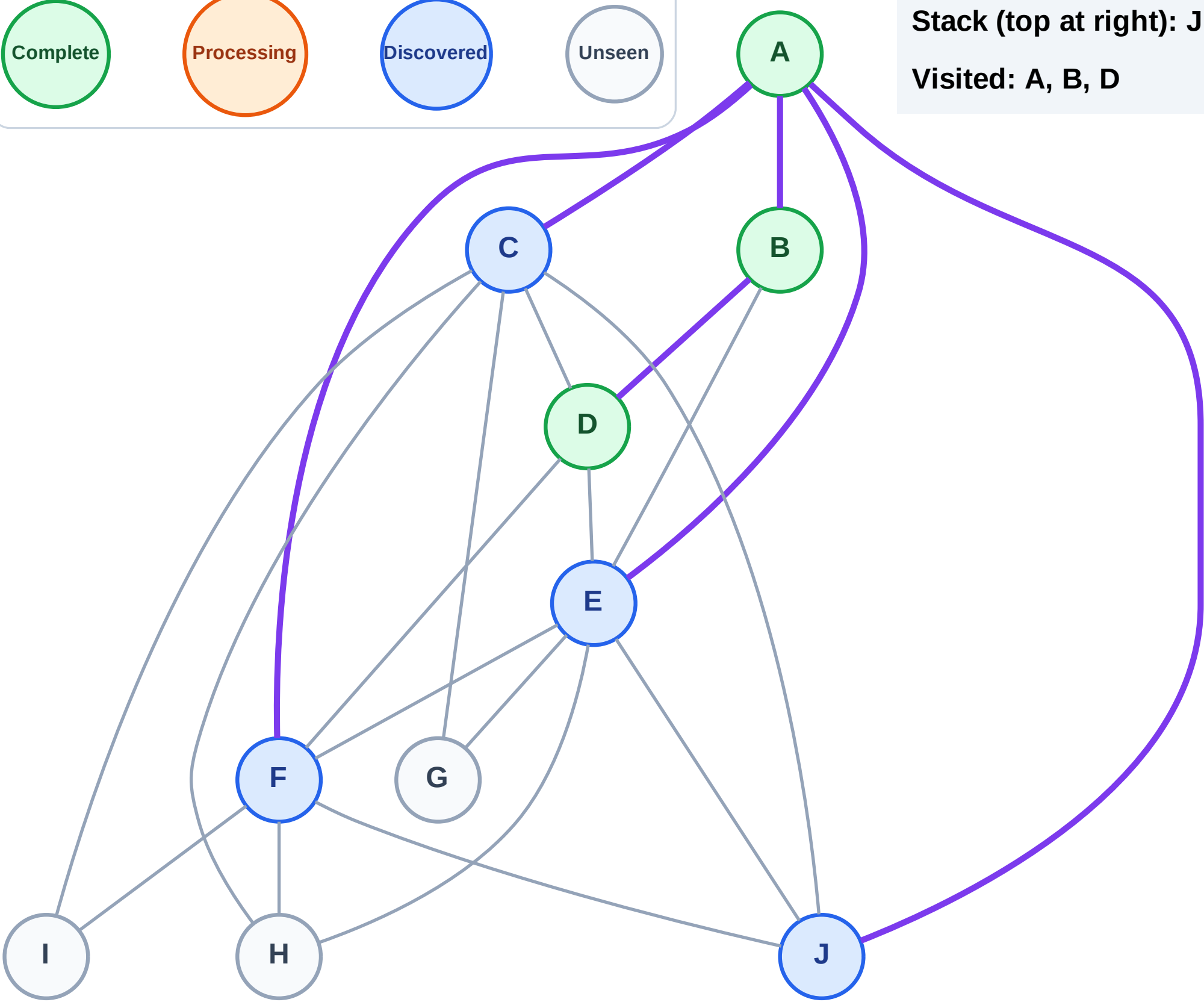
Complete

Processing

Discovered

Unseen

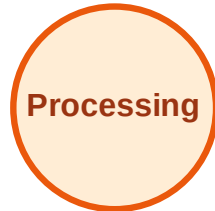
Stack (top at right): J | F | E | C
Visited: A, B, D



DFS Traversal

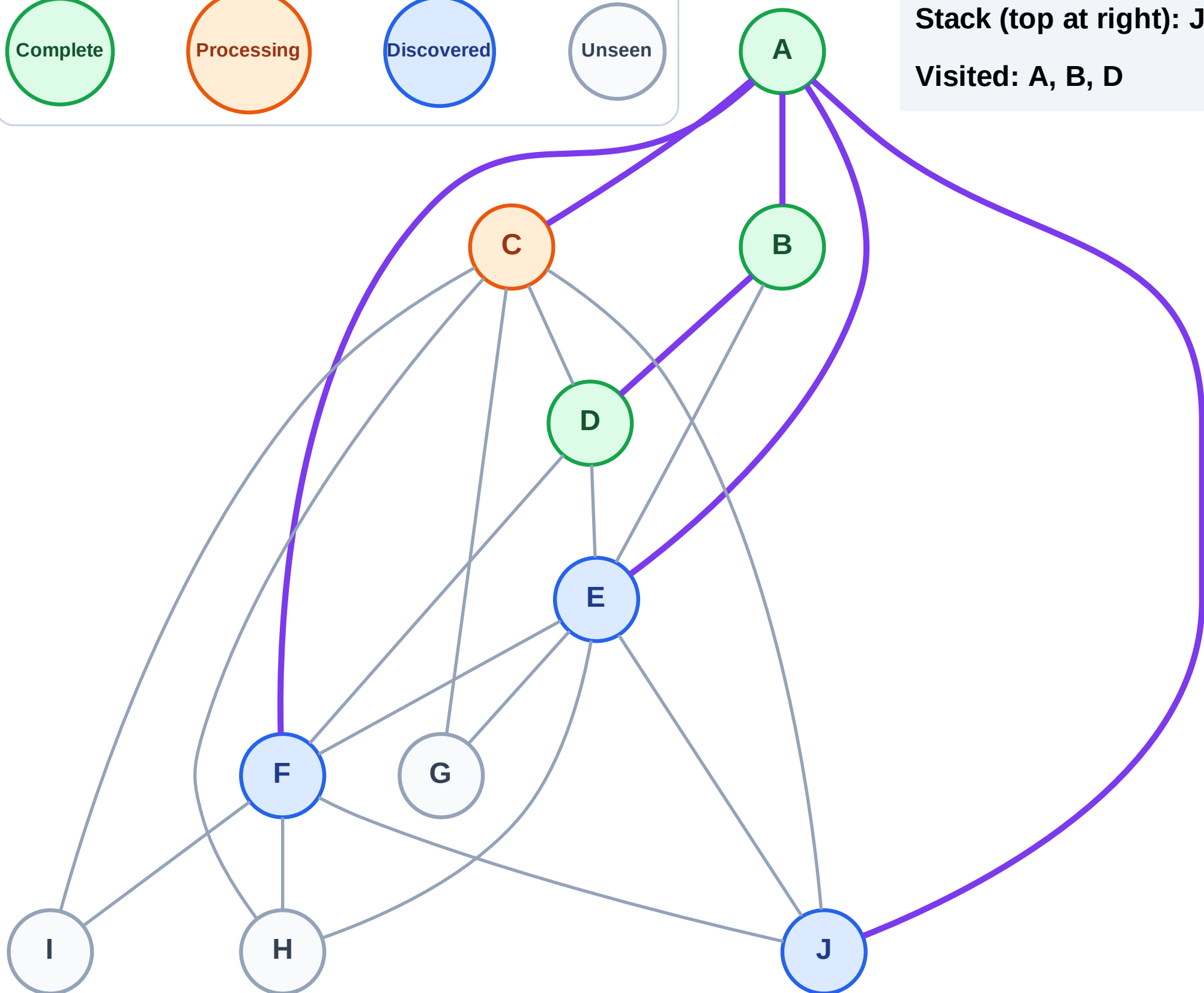
Step 8: Process node C

Legend



Stack (top at right): J | F | E

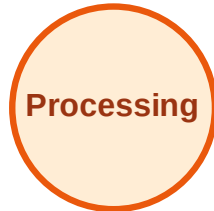
Visited: A, B, D



DFS Traversal

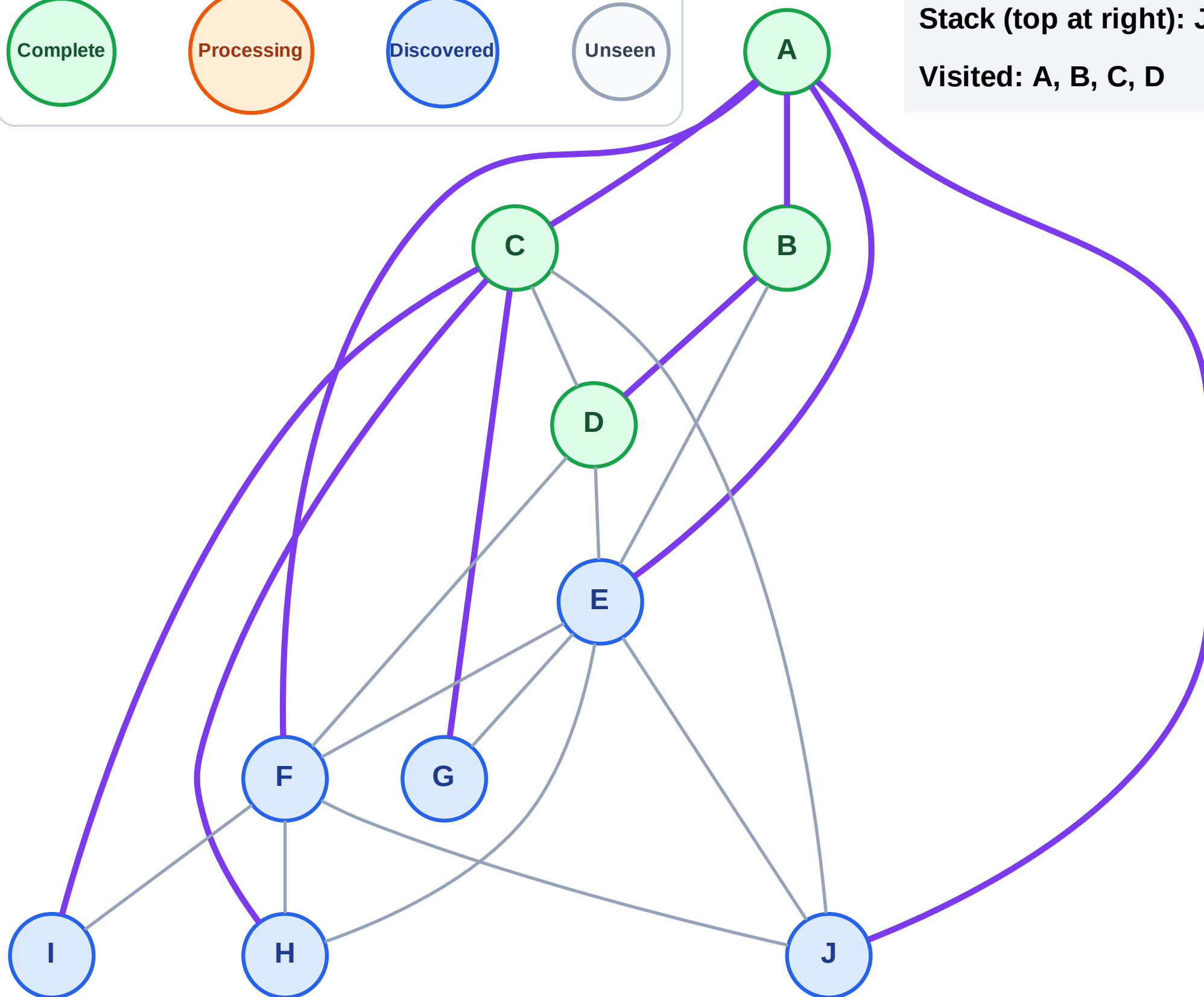
Step 9: Discover I, H, G from C

Legend



Stack (top at right): J | F | E | I | H | G

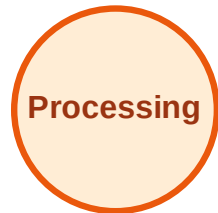
Visited: A, B, C, D



DFS Traversal

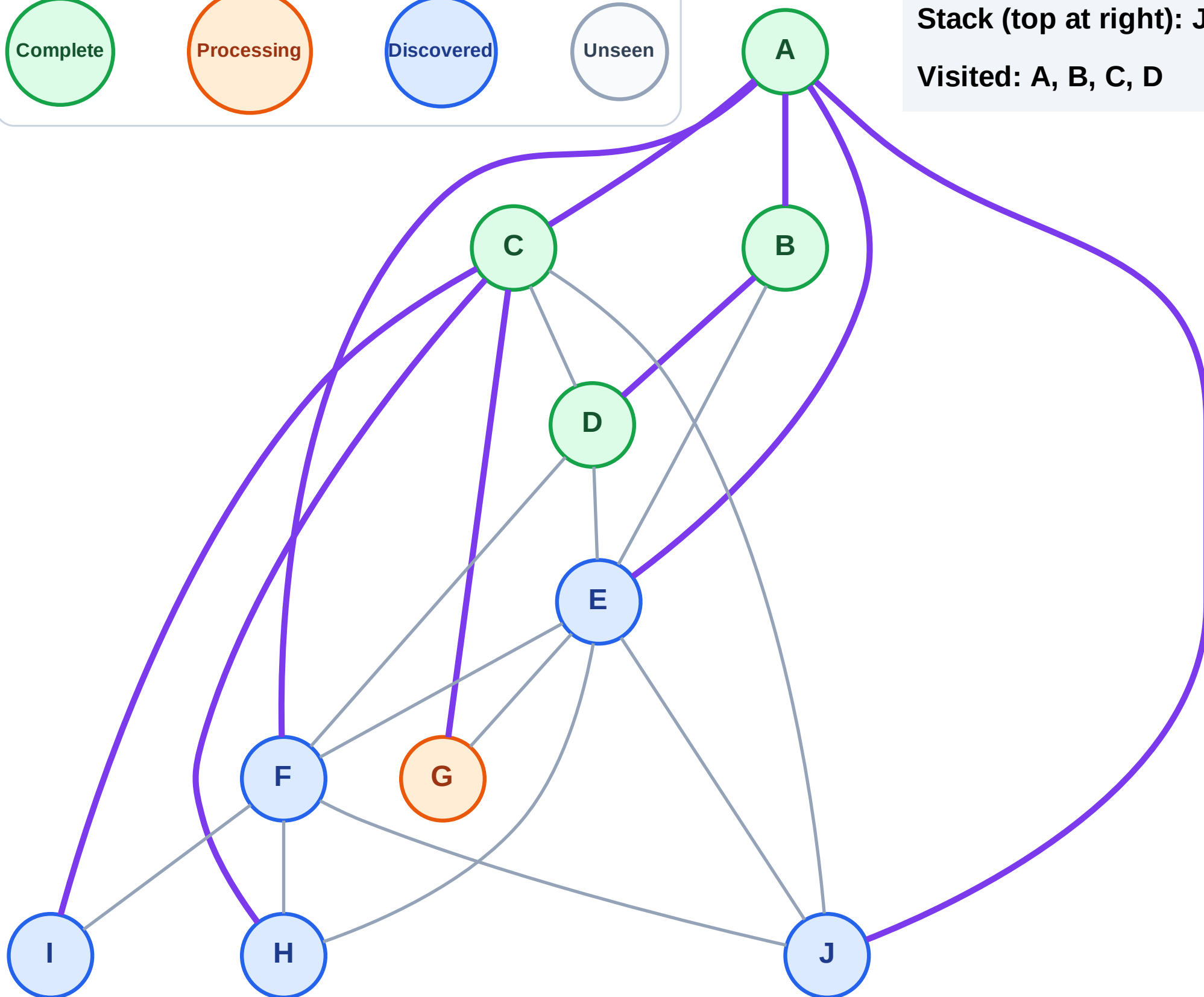
Step 10: Process node G

Legend



Stack (top at right): J | F | E | I | H

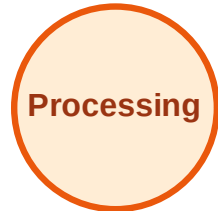
Visited: A, B, C, D



DFS Traversal

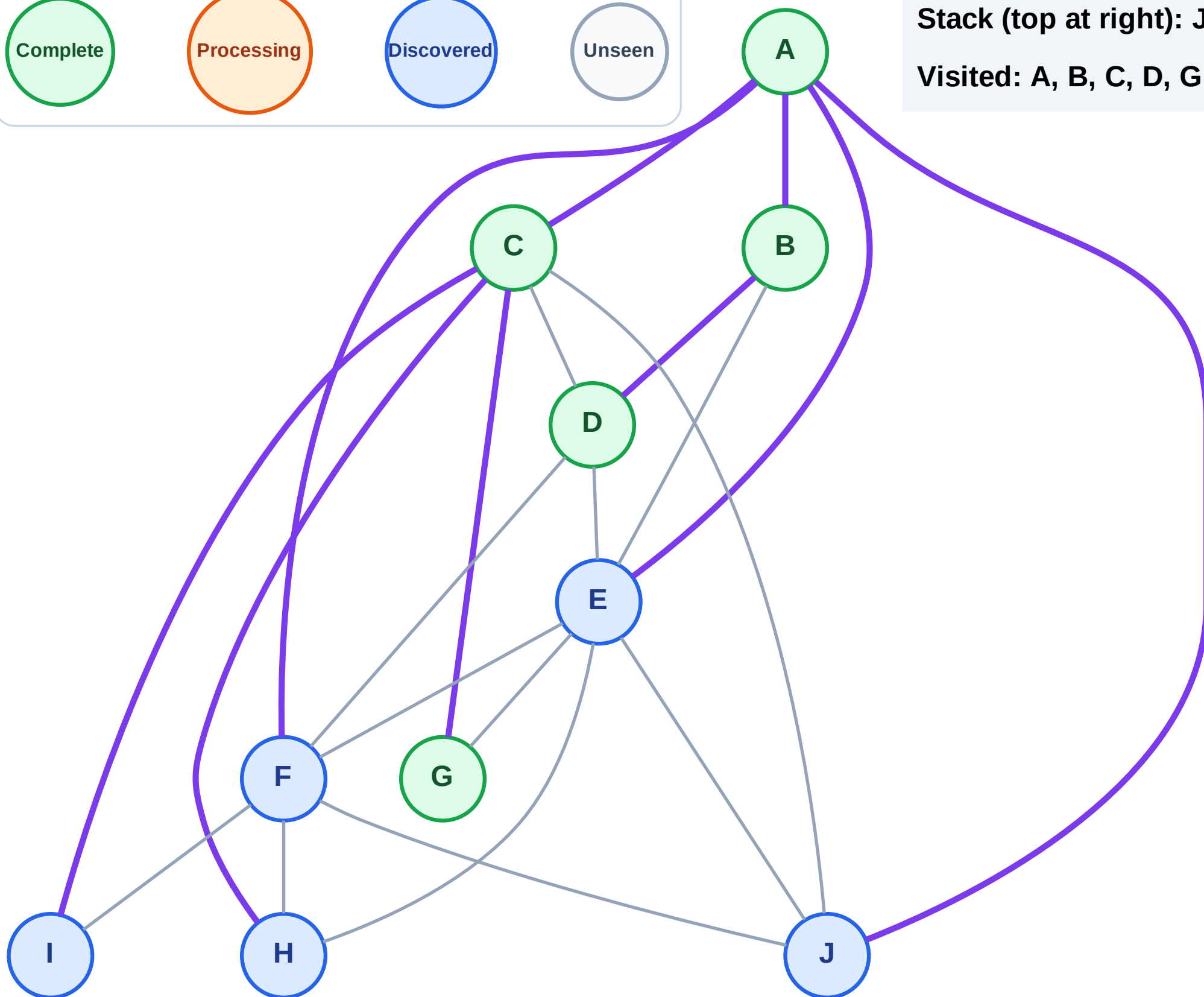
Step 11: No new neighbors from G

Legend



Stack (top at right): J | F | E | I | H

Visited: A, B, C, D, G



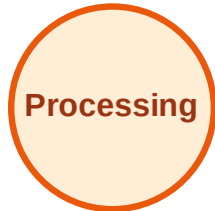
DFS Traversal

Step 12: Process node H

Legend



Complete



Processing



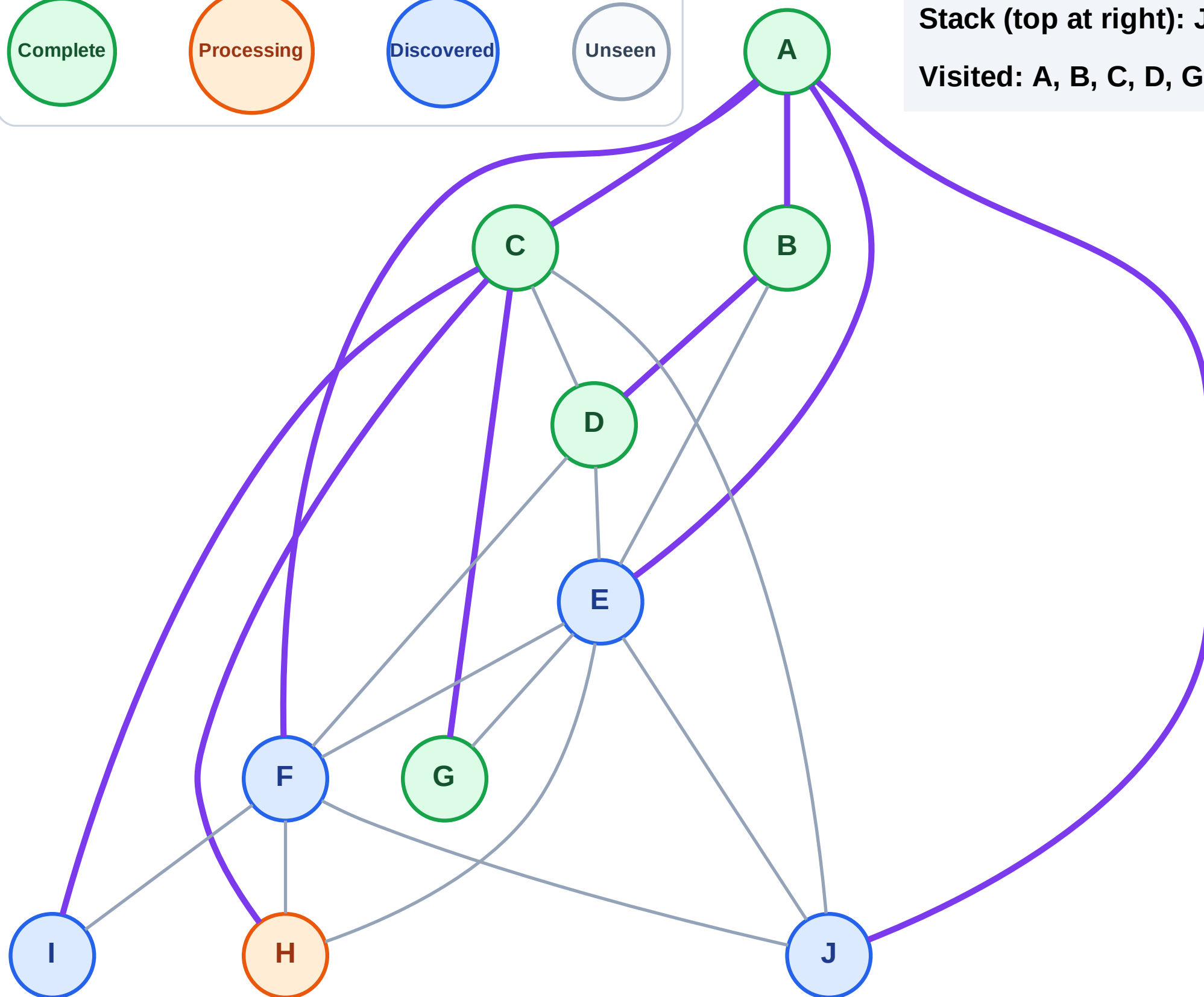
Discovered



Unseen

Stack (top at right): J | F | E | I

Visited: A, B, C, D, G



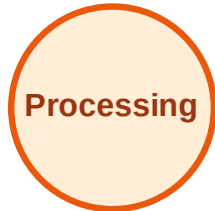
DFS Traversal

Step 13: No new neighbors from H

Legend



Complete



Processing



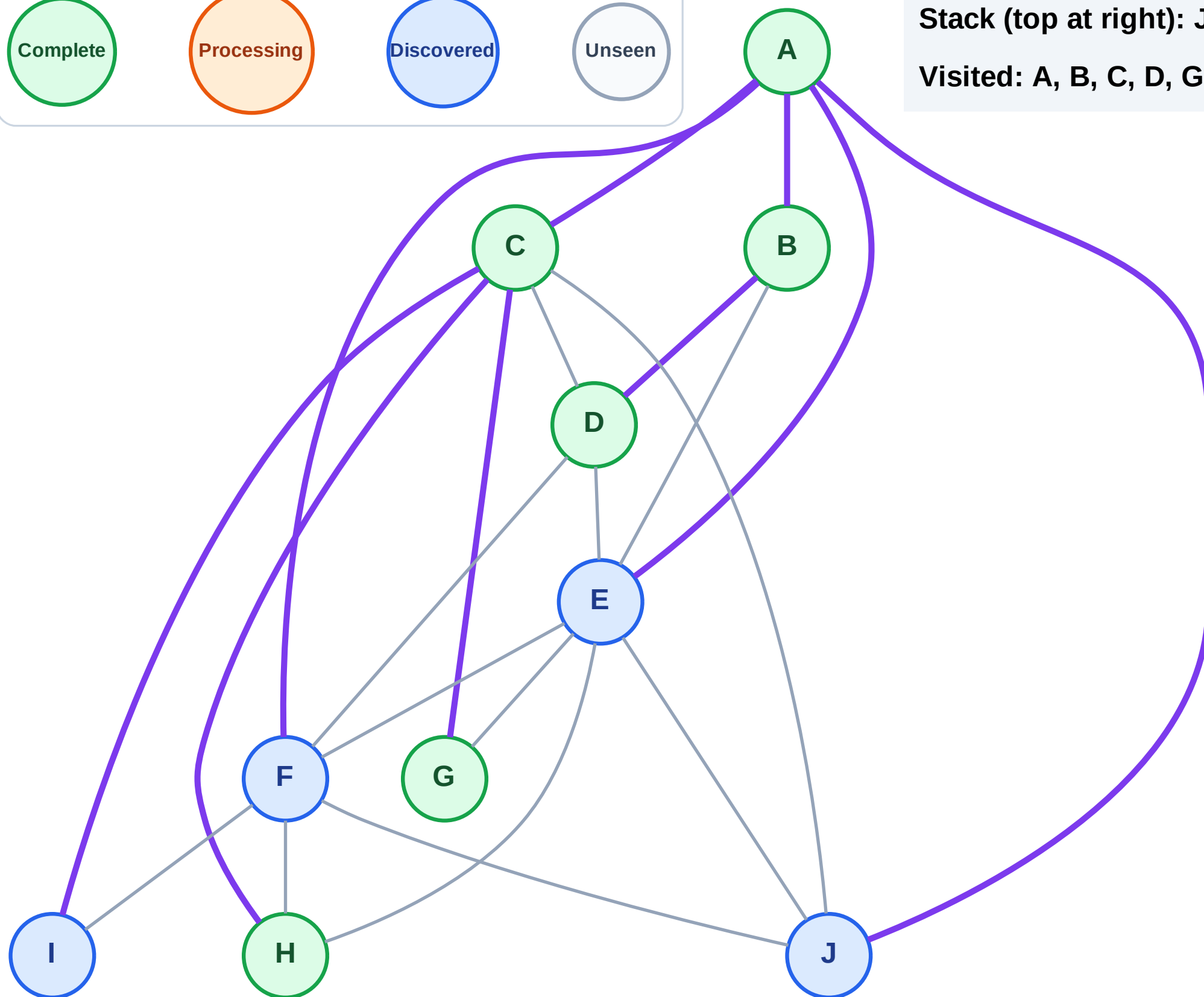
Discovered



Unseen

Stack (top at right): J | F | E | I

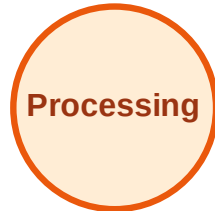
Visited: A, B, C, D, G, H



DFS Traversal

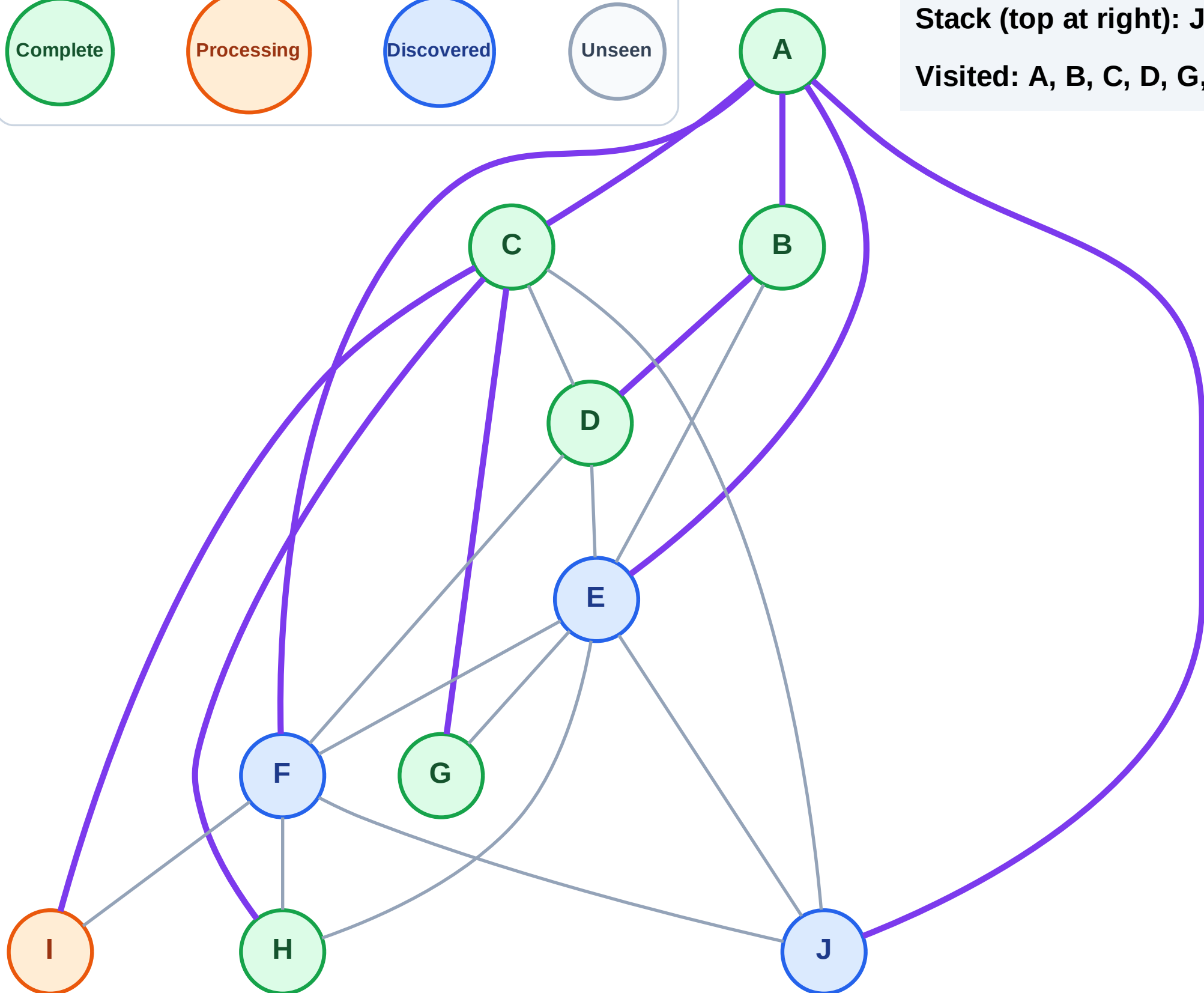
Step 14: Process node I

Legend



Stack (top at right): J | F | E

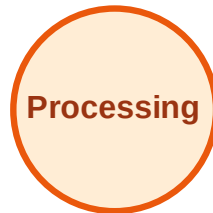
Visited: A, B, C, D, G, H



DFS Traversal

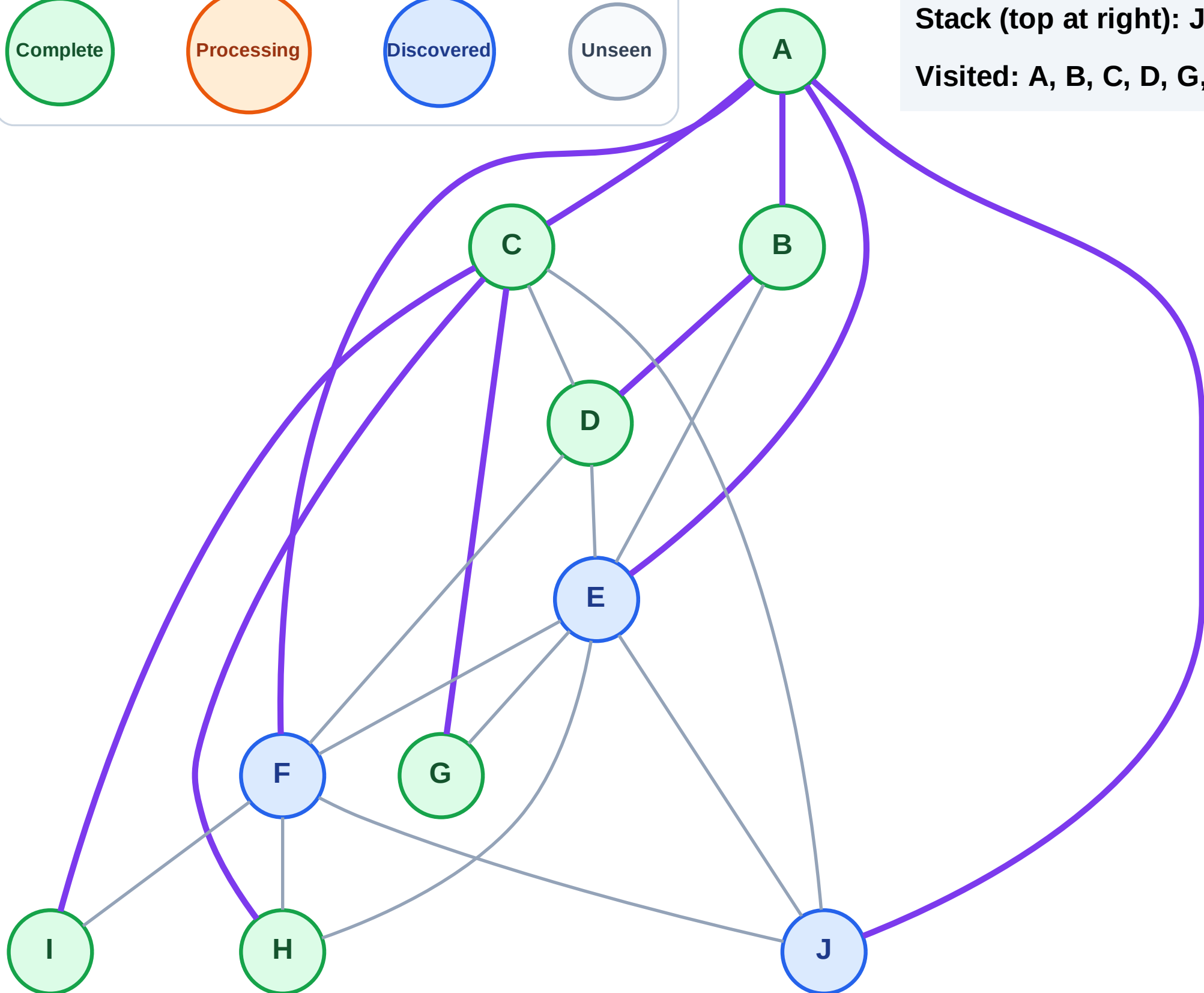
Step 15: No new neighbors from I

Legend



Stack (top at right): J | F | E

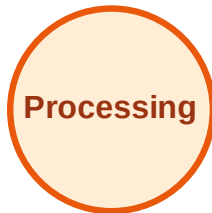
Visited: A, B, C, D, G, H, I



DFS Traversal

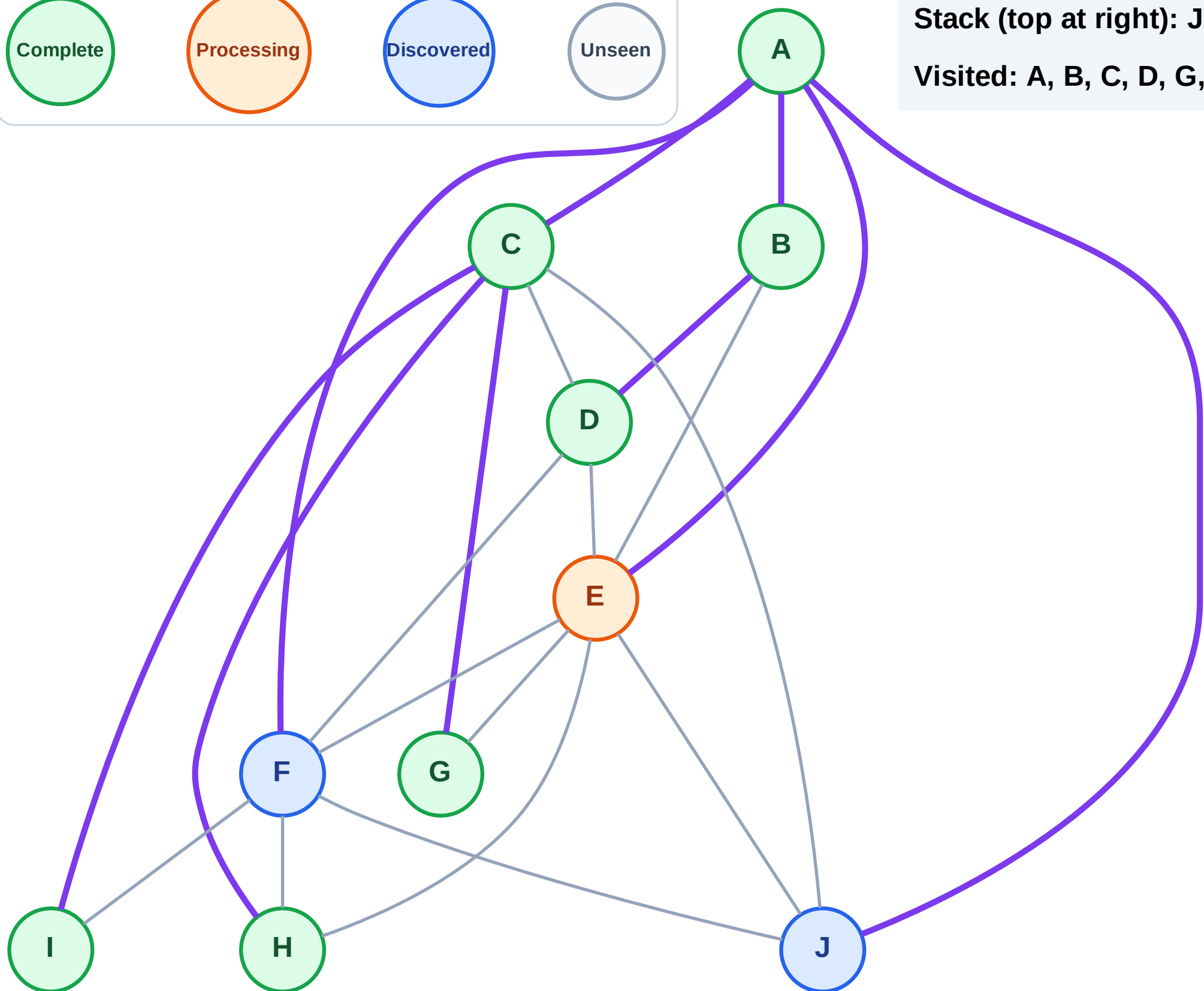
Step 16: Process node E

Legend



Stack (top at right): J | F

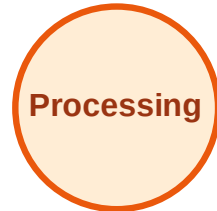
Visited: A, B, C, D, G, H, I



DFS Traversal

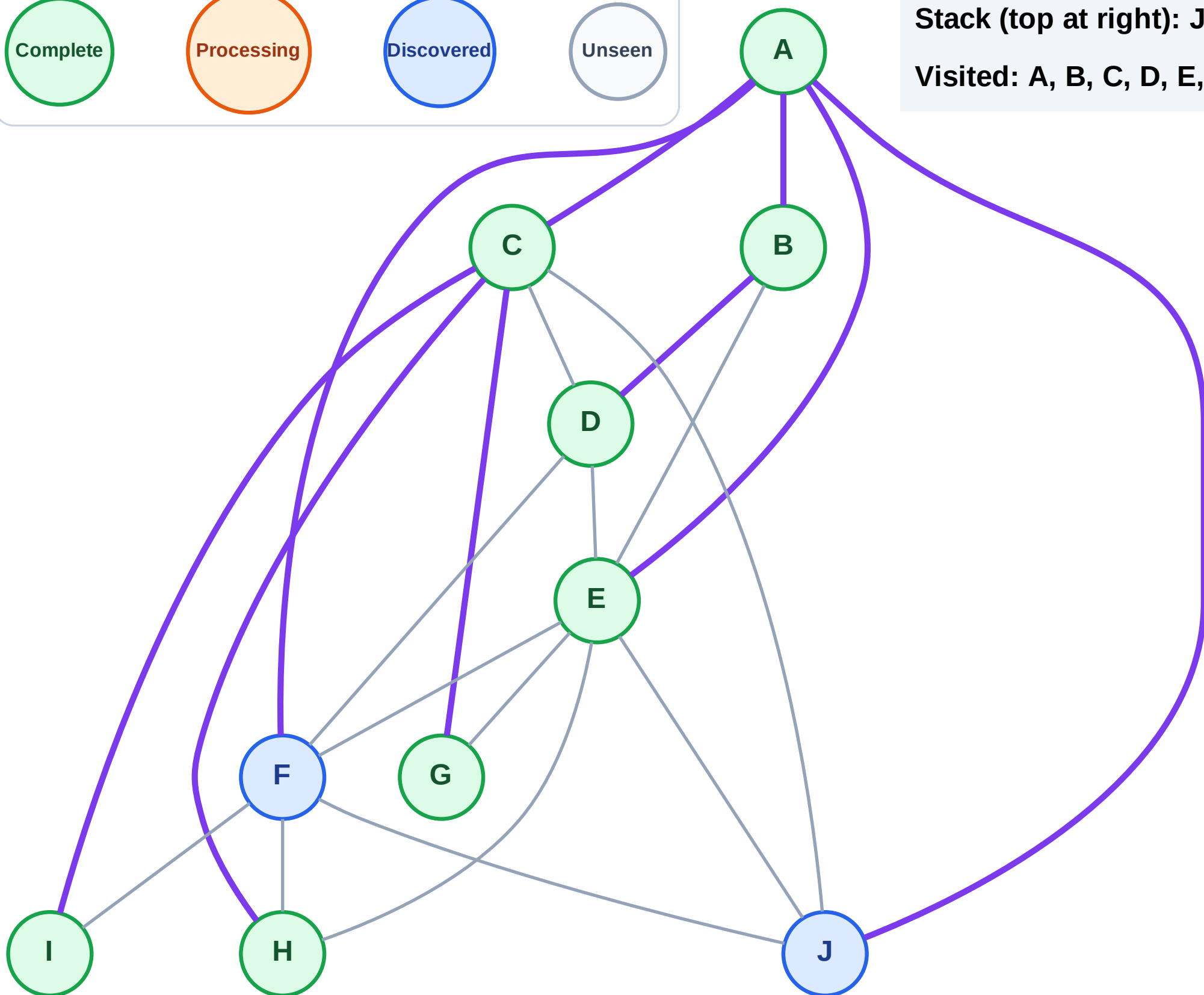
Step 17: No new neighbors from E

Legend



Stack (top at right): J | F

Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 18: Process node F

Legend

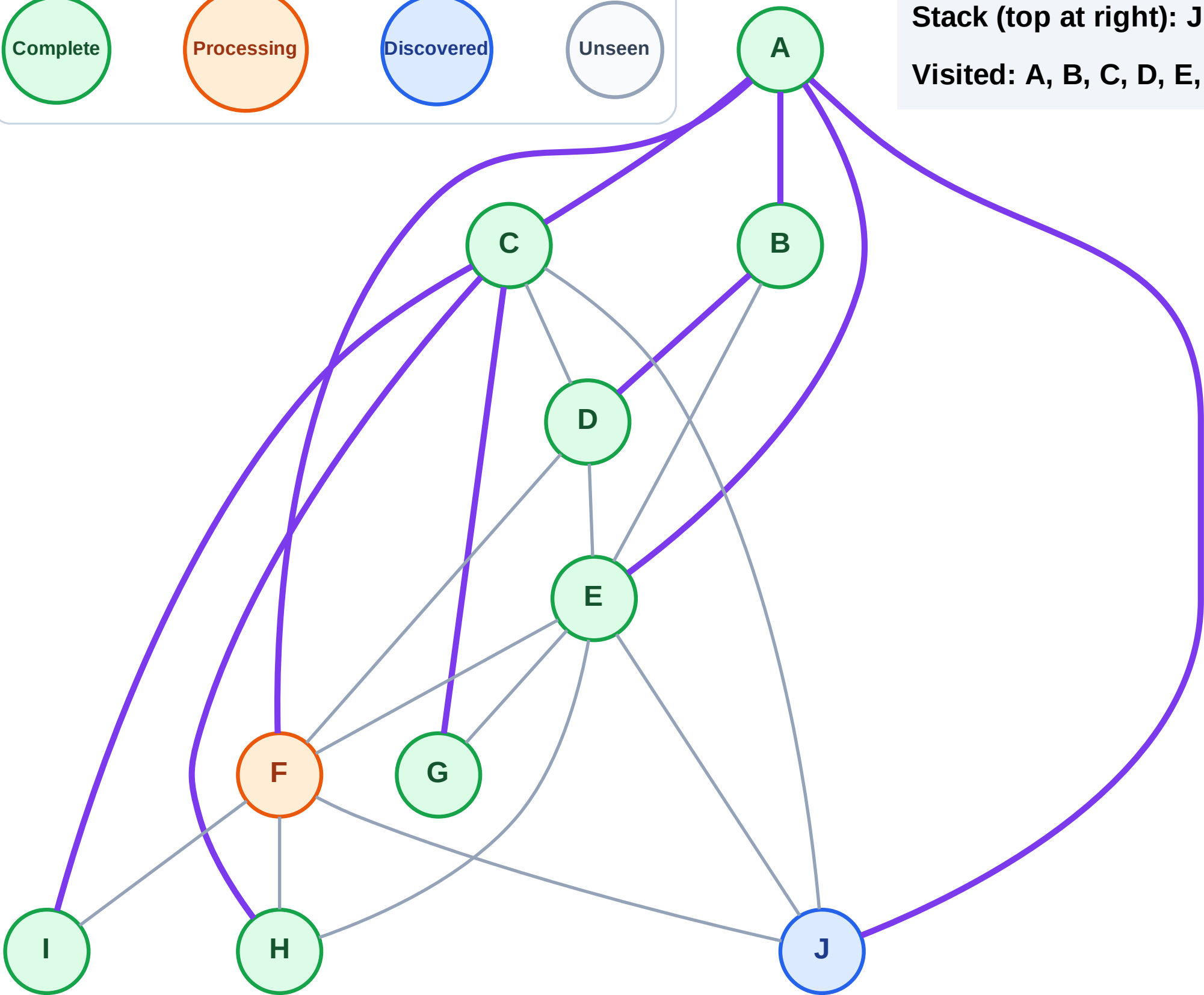
Complete

Processing

Discovered

Unseen

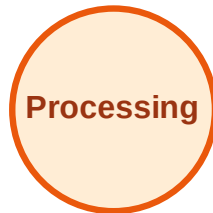
Stack (top at right): J
Visited: A, B, C, D, E, G, H, I



DFS Traversal

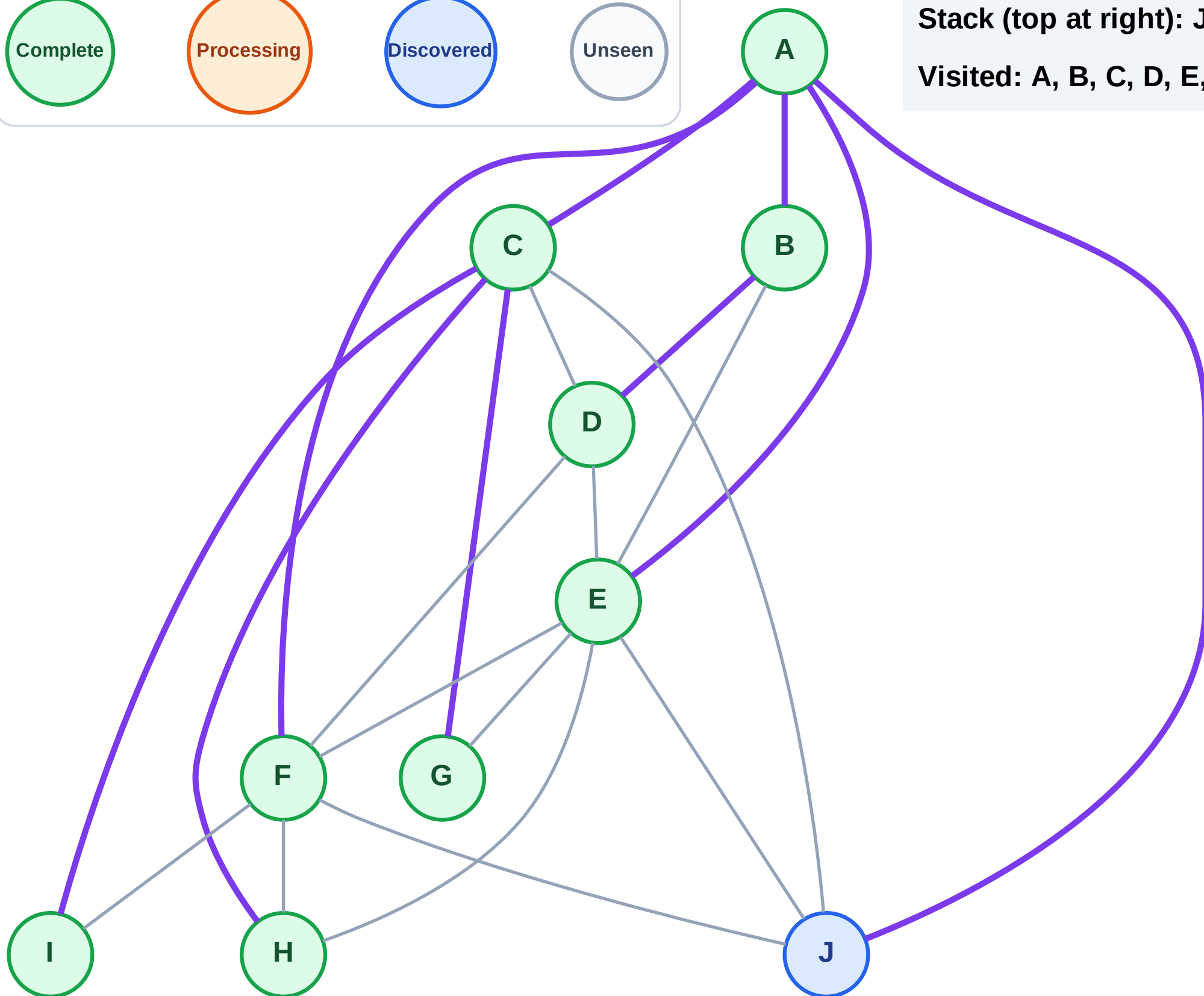
Step 19: No new neighbors from F

Legend



Stack (top at right): J

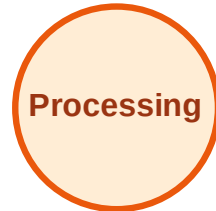
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

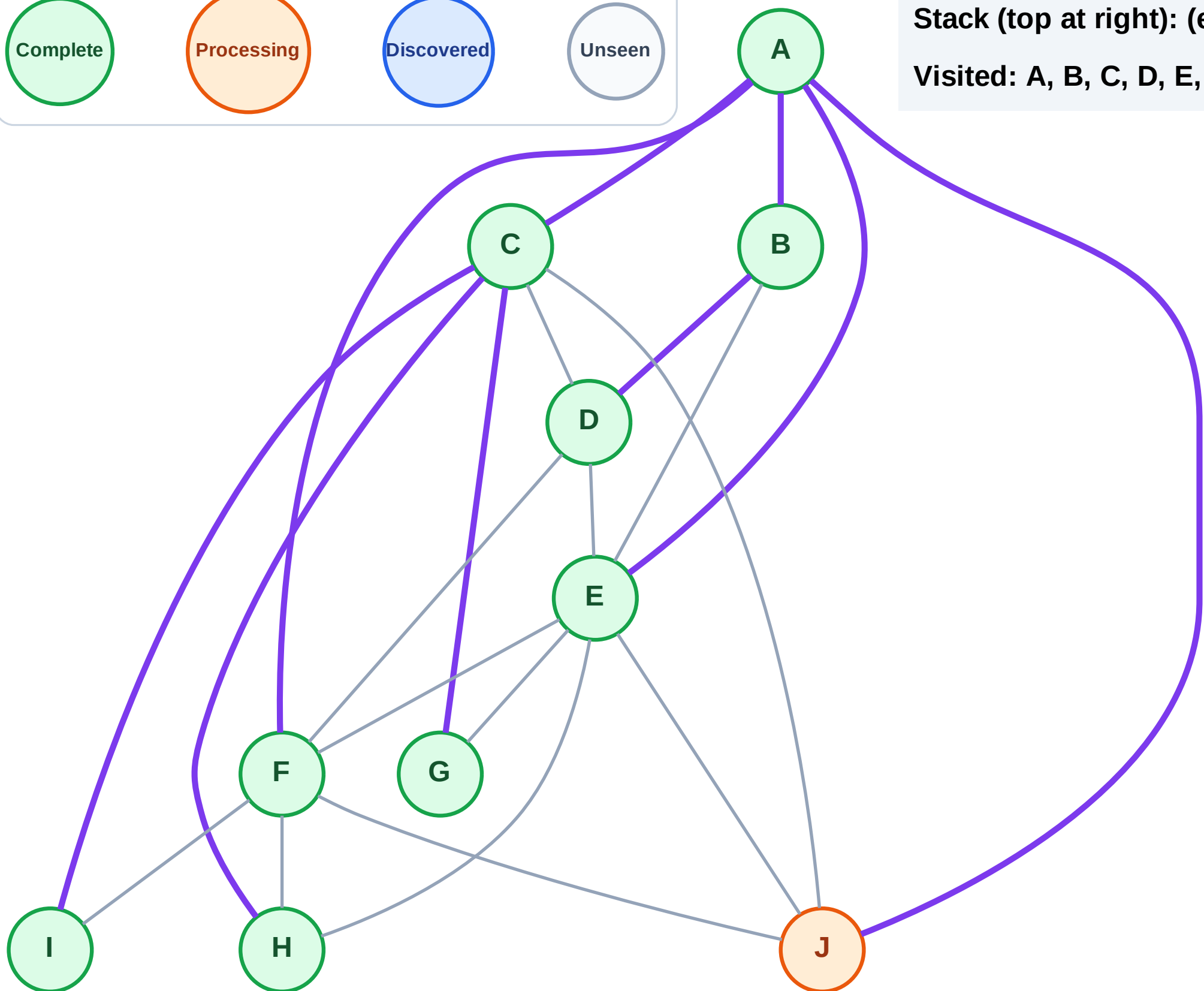
Step 20: Process node J

Legend



Stack (top at right): (empty)

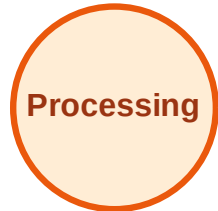
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

